

Iterative and Infrastructure-Aware Planning for Zero-Emission Vehicle Evacuations

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This work is supported by the California Climate Action Seed Grants of the University of California (Grant No. R02CP6996).

ABSTRACT

Efficient emergency evacuation of zero-emission vehicles (ZEVs) presents new challenges due to limited charging infrastructure, traffic congestion, and dynamic hazards. Designed explicitly as a macro-level strategic planning tool, this paper proposes a novel optimization framework that integrates traffic congestion, charging station queuing, and mobile charging station deployment into a unified model. Unlike existing approaches, it enables coordinated decision-making across all critical aspects of evacuation planning. The framework employs an iterative optimization strategy that prioritizes high-risk groups located close to hazards. This scheme plans routes based on allocated resources and remaining system capacity. The framework is evaluated using macroscopic network flow models and validated through high-fidelity microscopic simulations in SUMO on realistic disaster networks. Simulation results confirm this approach improves evacuation efficiency, reduces charging delays, and ensures timely support for high-risk populations. This solution provides a systematic tool for enhancing Zero-Emission Vehicle (ZEV) evacuation resilience under changing emergency conditions.

INDEX TERMS Optimization Problem, Electric Vehicle Evacuation, Facility-location-routing problem, Vehicle Routing Problem

I. Introduction

During emergency evacuations, ensuring the safety of communities with electric vehicles (EVs) presents several challenges. While both EVs and gasoline vehicles face increased traffic and infrastructure limitations during evacuations, EVs have specific issues due to their shorter range—approximately 60% of the range of gasoline vehicles [1]—and longer recharge times compared to gasoline vehicles' quick refueling [2]. Additionally, the availability of charging stations is less widespread, which becomes problematic in emergency situations. In 2023, there were over 53,000 public charging stations [3], compared to 118,678 convenience stores selling motor fuels [4], highlighting a significant gap. In addition to inherent range and charging limitations, EVs are also more vulnerable to power outages

that commonly accompany extreme weather events. Such outages can disable charging infrastructure, while sudden surges in electricity demand caused by large numbers of EVs attempting to charge simultaneously can further stress the already fragile grid [5] [6]. Addressing these vulnerabilities requires not only careful planning but also additional infrastructure support.

A. Planning of Evacuations

Figure 1 illustrates the key components of EV evacuation planning, each addressing a critical aspect of enabling safe and efficient vehicle movement under emergency conditions. These components include hazard prediction, departure scheduling, and destination assignment [7]–[9]; infrastructure impact evaluation on power, charging and road network

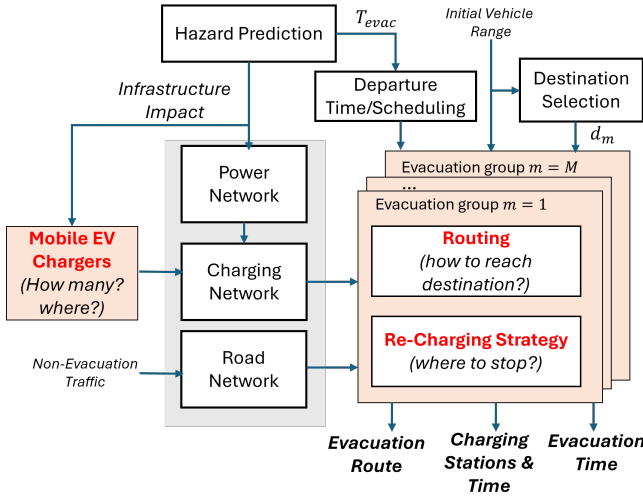


FIGURE 1: Overview of some of the key elements in the evacuation planning. Routing, Re-charging and deployment of mobile EV chargers are the key elements tackled in this work.

[6], [10]; coordinated routing and recharging strategies [11]–[18]; and the deployment of mobile charging stations [9], [12], [14], [16], [19], [20]. This paper focuses specifically on the latter two aspects: routing/recharging strategies and mobile charging station deployment, as they are central to ensuring evacuation feasibility for electric vehicles in the face of disrupted infrastructure and time-critical constraints. Much previous work has been done to address these challenges, both for daily-use oriented ([12], [14], [16], [17], [19], [21]–[23]) and emergency evacuation ([8], [9], [11], [13], [15], [18], [20]) scenarios.

1) Coordinated routing and recharging strategies

While macroscopic planning focuses on network-wide routing, recent studies address microscopic navigation, such as game-theoretic interactions [24], cognitive risk-integrated routing [25], and resilient cooperative control [26]. Complementing these, our work tackles the macro-level challenge of coordinating regional ZEV flows, queueing dynamics, and infrastructure placement under emergencies. Given the initial vehicle range and destination distance, the planning framework integrates the road, charging, and power networks while accounting for infrastructure impacts and background traffic, and makes two critical decisions: the routing strategy (how to reach the destination) and the recharging strategy (where and when to charge). Notable studies, such as [11], [12], [13], [14], [15], and [18], showcase how traffic-aware routing strategies can significantly enhance travel efficiency, particularly under time-critical or evacuation scenarios. For instance, [12] models drivers' equilibrium behavior in route and charging station choice, implicitly capturing traffic conditions as EV drivers seek to minimize overall travel time.

Similarly, [14] adopts stochastic user equilibrium constraints to model the impact of traffic congestion on EV path selection, enhancing the model's realism. [13] proposes a vehicle evacuation strategy for no-notice emergencies, using an optimization-simulation model that includes background traffic to simulate realistic traffic congestion. [11] proposes a capacity-constrained network flow model to maximize evacuees within a limited timeframe, utilizing an efficient Evacuation Scheduling Algorithm (ESA) for large, time-expanded networks. [15] iteratively updates travel times based on flow size, therefore optimizing the travel time under changing traffic conditions. In [18], a novel controllable equivalent circuit model was used to simulate traffic flow under limited road capacity.

Beyond routing, another critical aspect of ZEV evacuation planning is managing charging demand, especially under emergency conditions when large volumes of vehicles may rely on limited infrastructure. Several studies, including [12], [14], [16], [17], [27], incorporate charging capacity constraints or queuing dynamics at charging stations into their optimization frameworks. By accounting for delays caused by charger availability and demand surges, these approaches offer more realistic planning tools that help reduce bottlenecks, balance station utilization, and ensure smoother travel. [17] and [14] both address capacity-constrained energy infrastructure planning, which indirectly reduces queuing times at stations. The former focuses on a capacitated refueling station location and routing problem to prevent congestion, while the latter optimizes the placement of mobile charging stations (MCS) with explicit consideration of their capacity limits. [12] modeled drivers at each charging facility using an $M(t)/M/c$ queue to approximate average queuing time and waiting probability as functions of facility capacity and demand arrival rate. [16] estimates the waiting time at charging stations as a $M/M/c$ model. [27] further explores the joint optimization of transportation and power systems by evaluating time-varying EV charging pricing to manage grid loads and congestion during daily commutes. However, it is important to note that these works are primarily developed for routine, non-urgent driving scenarios, such as long-distance trips or daily commutes, and do not consider the unique time-critical and high-risk constraints of emergency evacuation contexts. This highlights a key research gap in extending charging queue models to urgent, large-scale evacuation planning for ZEVs.

2) Smart deployment of mobile charging stations

MCSs have emerged as a flexible and responsive solution for ZEV charging during emergency evacuations. Unlike fixed charging infrastructure, MCSs can be dynamically deployed along evacuation routes or in high-demand areas to provide on-the-go support where stationary infrastructure may be inadequate or compromised. Recent studies have explored optimization strategies for the placement and

deployment of both Fixed Charging Stations (FCSs) and MCSs to improve network performance during high-demand events. For example, [12] presents a model for locating FCSs that jointly considers infrastructure construction costs, driver travel time, and expected waiting times across the network. Similarly, [19] proposes a full-cover modeling framework to optimize FCS placement, particularly for long-distance travels. Building on this foundation, several works have focused specifically on MCS deployment strategies. [16] investigates the allocation and dispatch of MCSs to EV network hotspots during peak traffic hours to alleviate congestion. [14] introduces a multi-period optimization model to determine the optimal placement of MCSs over time. [28] explored the optimal deployment and assignment of Mobile Charging Stations (MCS) to maximize the total profit of the MCS operating agency while increasing the percentage of covered demand. In addition to normal traffic, [9] highlights the use of MCSs as a temporary supplement to FCSs during emergency operations, enhancing network redundancy and charging accessibility. [20] demonstrates that increasing the number of MCSs effectively reduces the evacuation time in both uniform and non-uniform EV parameter cases, highlighting the potential of MCSs to enhance evacuation efficiency.

While routing, queuing, and infrastructure deployment have received considerable attention, the integration of risk-based prioritization into evacuation planning remains largely underexplored. Evacuees may face varying levels of vulnerability due to proximity to hazards, mobility limitations, or lack of access to transportation. Addressing these disparities is essential for ensuring equitable and effective evacuation. Studies such as [8] and [9] have examined evacuation scheduling within defined time windows. However, only a few, such as [11], have introduced frameworks that assign region-specific risk weights to prioritize evacuation efforts. This gap highlights the need for future research to incorporate social vulnerability and risk assessment into optimization models for ZEV evacuations.

B. Optimization Approaches

To address the challenges in evacuation planning, researchers have focused on various types of optimization problems and their corresponding solutions. As optimization problems increase in complexity due to the expansion of graph size and the addition of more constraints, finding efficient solutions becomes more difficult. In response to these challenges, various approaches have been developed in the literature.

A commonly employed strategy in optimization is the use of heuristic methods. For example, [16] addressed the management of mobile charging stations by modeling it as a quadratic assignment problem (QAP) and applying ant colony optimization (ACO) to derive a meta-heuristic solution. [8] developed a heuristic solution to address the complexity of Vehicle Routing Problem (VRP) based problem. [21] proposed a new heuristic solution to optimize

the charging schedules for the generated routes. In [15], the authors introduced a route planning method that uses spanning trees to model the routing process. Additionally, a branch-and-price algorithm was employed to optimize travel and refueling times in a realistic network. Similarly, [17] developed two optimization methods to address the increased complexity of solving the capacitated refueling station location problem with routing. The first method builds upon an existing branch-and-cut approach, while the second is a branch-cut-and-price algorithm that incorporates variables corresponding to feasible routes. [11] uses a greedy algorithm to compute the maximum flow along each identified route, enabling the determination of optimal evacuation schedules for each time interval.

Another approach involves the reformulation of nonlinear optimization problems into linear ones. In [14], a multi-period nonlinear flow-refueling location optimization problem was transformed into a linear integer programming model, facilitating more efficient solutions. In [22], the authors tackled the location-allocation problem by transforming a nonlinear model, which incorporated queuing constraints, into an equivalent mixed-integer linear optimization problem. This reformulation leveraged the computational advantages of linear optimization techniques while maintaining the complexity of resource allocation under congestion. Consequently, this approach allowed for more efficient problem-solving, even in the context of large-scale networks with high demand variability.

Decomposing optimization problems into multiple layers is also an effective technique for reducing computational complexity. The study by [23] first proposed a problem of determining when and how to refuel, given that the routing decision has already been made, and then compared it with the integrated problem of routing and refueling, showing the latter problem is more complex and difficult to solve. [9], [20] and [11] applied Dijkstra's algorithm to identify optimal evacuation routes, treating the remaining challenges, such as scheduling and vehicle grouping, as optimization problems. Similarly, [21] divided the capacitated EV routing problem into two subproblems—vehicle routing and fixed-route charging—to enable a more efficient solution. [13] employed a bilevel model to optimize departure times, routes, and destinations in the upper level, using the lower level to reallocate resources and eliminate bottlenecks. This layered approach is also seen in [12], where an upper-level model determines the optimal location and capacity of facilities, while a lower-level model predicts drivers' route and charging choices. Similarly, [19] proposed two optimization models: one to minimize the total cost of station placement and en-route recharging, and another to optimize the placement of a fixed number of stations to minimize recharging requirements.

Table 1 summarizes representative studies, highlighting key features such as evacuation, traffic-aware routing, queuing models, charging station deployment, and prioritization

of high-risk groups. It also catalogs their optimization problem formulations and solution methodologies. Although each of these studies addresses an important aspect of ZEV evacuation, they typically treat these factors in isolation. Most routing-based works neglect energy and queuing constraints, leading to evacuation plans that are infeasible for limited-range EVs under charging bottlenecks. Conversely, studies that incorporate charging dynamics often omit evacuation-specific time-critical constraints or traffic interactions, resulting in unrealistic or overly idealized solutions. Moreover, existing formulations generally lack mechanisms for adaptive prioritization of vulnerable zones, assuming all evacuees have equal urgency. The absence of a unified framework to jointly consider traffic congestion, energy limitations, charging capacity, and risk-based prioritization leads to fragmented and suboptimal evacuation strategies (e.g., congestion at chargers or delayed aid to high-risk groups). To address these limitations, this paper makes the following key contributions:

- **Prioritized High-Risk Evacuation for ZEVs:** We developed a unique model that prioritizes high-risk groups located closer to hazards by employing an iterative optimization approach. In contrast to the method proposed in [11], which uses region-specific risk weights to influence evacuation priorities, our approach recursively plans evacuation routes by updating decisions at each stage based on prior resource allocations and remaining system capacity. This results in a more adaptive and dynamic prioritization strategy that better captures the evolving nature of emergency scenarios.
- **Integrated and Enhanced Multi-Factor EV Evacuation Optimization:** This paper proposes and solves a novel mixed-integer linear programming optimization problem that comprehensively integrates ZEV evacuation routing and charging planning with real-world factors such as traffic congestion, charging station queuing, and mobile charging station placement. Building upon the foundational model in [23], we significantly enhance its applicability to emergency scenarios by incorporating these additional constraints and infrastructure considerations. This holistic and extended formulation addresses key limitations in prior literature, which often treat these components in isolation or as secondary factors.

It is important to note that this framework is positioned as a strategic, macro-level planning model used to evaluate resource allocation and pre-position infrastructure prior to an event, rather than as a real-time operational tool. Throughout this paper, while we use the broader term ZEV in our title and framing to align with clean-transportation policy, our specific mathematical formulation and analysis focus exclusively on battery-electric vehicles (BEVs). However, the proposed optimization framework is flexible and can be naturally extended to hydrogen fuel-cell electric vehicles,

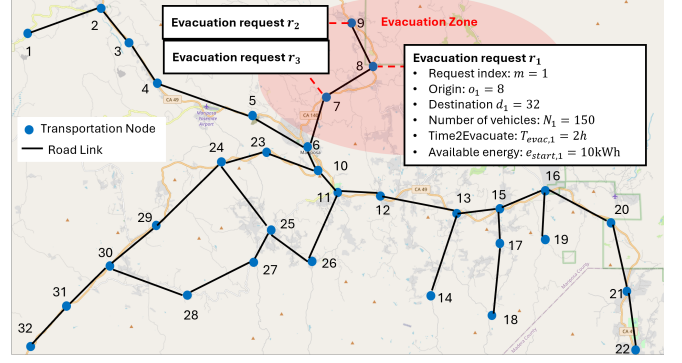


FIGURE 2: Illustration of an evacuation request r_m and transportation network.

which face analogous range and refueling constraints during emergencies.

II. Modeling and Problem Formulation

This section introduces the notation that will be used in this work.

A. Transportation Flow Model

The transportation network is modeled using a graph as illustrated in Figure 2, composed of nodes $\mathcal{V}_T \subset \mathbb{N}$ and links $\mathcal{L}_T \subset \mathcal{V}_T \times \mathcal{V}_T$. The nodes capture intersections $i \in \mathcal{V}_T$, which are connected by road segments or links $(i, j) \in \mathcal{L}_T$. Inspired by [23], we introduce evacuation requests as follows:

Definition 1. (Evacuation Request) An evacuation request r is a tuple $r = (o, d, N, T_{evac}, e_{start}, E_{max}, g) \in \mathcal{V}_T \times \mathcal{V}_T \times \mathbb{R}_+^4 \times \mathcal{G}$, where $o \in \mathcal{O} \subset \mathcal{V}_T$ is the origin of the evacuation, $d \in \mathcal{D} \subset \mathcal{V}_T$ is the destination, N is the number of vehicles that are expected to travel between o and d , T_{evac} is the time window to evacuate. The variable e_{start} represents the available energy of each vehicle at the beginning of the journey, E_{max} is the maximum energy that can be stored in the vehicle and $g(s) : \mathbb{R} \rightarrow \mathbb{R}$ is the energy consumption map (in kWh/h) that is possibly speed (s) dependent. The set of all evacuation requests is defined as $\mathcal{R} = \{r_m\}_{m \in \mathcal{M}}$ where $\mathcal{M} = \{1, \dots, M\}$ and M is the total number of requests, $\mathcal{O}$ the set of all origins, $\mathcal{D}$ the set of all destinations and $\mathcal{G}$ the set of all energy consumption maps.

Emergency managers may need to consider multiple evacuation requests. $\mathcal{O}$ represents the set of neighborhoods or zones that need to be evacuated due to the imminent arrival of a natural disaster. The set $\mathcal{D}$ is the set of shelters, which is usually pre-designed by the local authorities [29]. The number of vehicles that will be evacuated N can be determined using survey data or traffic observation for a given area [15], [30]. T_{evac} can be estimated based on the expected rate of propagation of the natural hazard [8]. The initial energy that each vehicle has for the evacuation (e_{start})

TABLE 1: Previous works on optimization of ZEV evacuation planning with charging stations

Reference	Evacuation	Routing with Traffic Aware	Queuing Model	Charging Station Deployment	Prioritization for high-risk groups	Problem Type	Solution Approach
[11]	✓	✓	-	✗	✓	Integer Programming	Decomposition, heuristic
[16]	✗	✗	$M/M/c$	✓	✗	Quadratic Assignment	Heuristic
[12]	✗	✓	$M(t)/M/c$	✓	✗	Nonlinear Programming	Decomposition
[13]	✓	✓	-	✗	✗	Linear Programming	Decomposition
[17]	✗	✗	-	✗	✗	Integer Programming	Heuristic
[21]	✗	✗	-	✗	✗	Nonlinear programming	Decomposition, heuristic
[15]	✓	✓	-	✗	✗	Mixed-integer Nonlinear Programming	Heuristic
[19]	✗	✗	-	✓	✗	Linear Programming	Decomposition
[14]	✗	✓	-	✓	✗	Linear Integer Programming	Linearization
[22]	✗	✗	-	✓	✗	Mixed Linear Programming	Linearization
[8]	✓	✗	-	✗	✗	Mixed-integer Linear Programming	Heuristic
[9]	✓	✗	-	✓	✗	Mixed Integer Linear Programming	Decomposition
[23]	✗	✗	-	✗	✗	Mixed Integer Linear Programming	Decomposition
[20]	✓	✗	-	✓	✗	Mixed Integer Linear Programming	Decomposition
[18]	✓	✓	-	✗	✗	Mixed Integer Linear Programming	Linearization
Our work	✓	✓	$M/M/c$	✓	✓	Mixed-integer Linear Programming	Linearization

is a variable that has a high level of uncertainty and it is difficult to know in advance. The maximum amount of energy that can be stored in the (electric) vehicle E_{max} as well as the energy consumption are other factors that often are not exactly known to evacuation planning agencies.

Based on the evacuation request, we can compute the average flow of vehicles traversing $o_m - d_m$ as follows:

$$\alpha_m = \frac{N_m}{T_{evac,m}}, \quad m \in \mathcal{M} \quad (1)$$

where α_m is the expected traffic flow (in cars/h).

Assumption 1. *There is at least one path¹ connecting the nodes o_m and d_m for all $m \in \mathcal{M}$.*

This assumption ensures problem feasibility. In realistic evacuation networks, disconnected zones could be handled through separate local plans or airlift. Therefore, assuming at least one available path aligns with real-world planning practice and keeps the optimization problem well-posed.

¹A path is defined as an ordered sequence of nodes such that any pair of consecutive nodes represents a link in $\mathcal{L}_T$

Assumption 2. *Traffic in the transportation network operates in steady state conditions.*

This abstraction ensures analytical tractability and allows the model to capture key structural properties of network capacity, bottleneck formation, and demand distribution. Although evacuations are dynamic, this approach models the average flow at equilibrium, which is a common practice in transportation planning [31], [32]. A fully dynamic model would add significant complexity without substantially improving insights for routing and charging deployment at the planning stage. However, by relying on steady-state equilibrium, this model likely underestimates peak transient congestion, such as temporary spikes caused by sudden merge bottlenecks, while potentially overestimating total system clearance times, as it cannot account for dynamic, on-the-fly rerouting by individual drivers as queues build up.

We denote f_{ij}^m as the (steady) vehicle flow (cars per hour) that traverse the road link $(i, j) \in \mathcal{L}_T$ due to the evacuation request m . The overall flow of vehicles in each road link is

given by:

$$f_{ij} = \beta_{ij} + \sum_{m \in \mathcal{M}} f_{ij}^m \quad (2)$$

where β_{ij} is pre-evacuation traffic flow. To guarantee conservation of vehicles in the traffic nodes, the total number of flows entering a given node needs to match the outgoing flow. This can be cast as:

$$\sum_{i \in \mathcal{V}^+(k)} f_{ik}^m = \sum_{j \in \mathcal{V}^-(k)} f_{kj}^m + (\mathbf{1}_{k=d_m} - \mathbf{1}_{k=o_m})\alpha_m, \quad (3)$$

$$k \in \mathcal{V}_T, m \in \mathcal{M}$$

where $\mathcal{V}^+(k)$ represents the set of nodes that have an incoming connection with node k and $\mathcal{V}^-(k)$ the set of outgoing nodes from k . The function $\mathbf{1}_{k=m}$ is an indicator function that yields 1 if $k = m$ and zero otherwise. The second term on the right side of the equation captures the source/sink of vehicles that will be injected/removed from the transportation network for each request r_m . In this work we assume that the flow in each link is non-negative:

$$f_{ik}^m \geq 0 \quad (i, k) \in \mathcal{L}_T, m \in \mathcal{M} \quad (4)$$

B. Travel Time

The road link ij has a length d_{ij} and an average traveling speed (without congestion) of s_{ij}^{ff} . Under this "free flow" assumption, the travel time is $t_{ij}^{ff} = d_{ij}/s_{ij}^{ff}$. To model traffic congestion, we consider the Bureau of Public Roads (BPR) function [33], which models the increase in travel time as a function of the traffic flow in the link:

$$t_{ij}(f_{ij}) = t_{ij}^{ff} \left(1 + b_1 \left(\frac{f_{ij}}{f_{ij}^{ff}} \right)^{b_2} \right) \quad (5)$$

where $b_1 \geq 0$ and $b_2 > 0$ are parameters that determine the sensitivity of travel time to changes in traffic volume. $f_{ij}^{ff} > 0$ is the free-flow limit before congestion starts to emerge on the road segment ij .

Assumption 3. All vehicles in a road segment $(i, j) \in \mathcal{L}_T$ travel with the same speed ($s_{ij} = d_{ij}/t_{ij}(f_{ij})$).

While individual vehicle speeds naturally vary in reality, assuming a uniform speed for a given traffic density is a well-established and necessary simplification in macroscopic traffic flow modeling [33].

The overall driving time for the evacuation request r_m is defined as:

$$T_{drive}^m = \sum_{(i,j) \in \mathcal{L}_T} t_{ij}(f_{ij}) \mathbf{1}_{\{f_{ij}^m > 0\}}, \quad m \in \mathcal{M} \quad (6)$$

This metric evaluates the overall driving time along the road links used by vehicles N_m associated with request r_m . If this group of vehicles uses a single path, then T_{drive}^m capture the evacuation time. If multiple paths are deployed, then T_{drive}^m provides an upper bound estimation for the evacuation time for request r_m .

TABLE 2: List of Key Parameters of the Evacuation Problem

Symbol	Description
Evacuation requests	
$\mathcal{M}$	List of evacuation requests;
$\mathcal{O}$	List of origins
$\mathcal{D}$	List of destinations
o_m	origin associated with request m
d_m	destination associated with request m
α_m	flow of vehicles associated with request m [cars/h]
Transportation Parameters	
$\mathcal{V}_T$	set of transportation nodes
$\mathcal{L}_T$	set of transportation links ($\subset \mathcal{V}_T \times \mathcal{V}_T$)
d_{ij}	distance between node i and j [mile]
f_{ij}^{ff}	free flow on link ij [cars/h]
s_{ij}^{ff}	average traveling speed on link ij without congestion [mph]
β_{ij}	pre-evacuation traffic flow on link ij [cars/h]
b_1	BPR function parameter
b_2	BPR function parameter
Energy and Charging Parameters	
c_j	number of charging ports at node j
$p_{ch,j}$	charging power
e_{start}^m	initial energy available in the vehicle [kWh]
E_{max}^m	the maximum energy that the vehicle can hold [kWh]
$g_m(s_{ij}^{ff})$	energy consumption per unit distance [kWh/mile] for the vehicle fleet N_m (at free flow speed)
t_{setup}	setup time to access the charging station
$t_{ch,max}$	max charging time at charging station
t_{queue}^{max}	max queuing time at charging station
n_{MCS}	total number of MCS

C. Routing Problem (RP)

In this work, we consider a weighted average evacuation time among all the evacuation requests as the cost function:

$$J_R = \frac{1}{M} \sum_{m \in \mathcal{M}} w_{request}^m T_{drive}^m \quad (7)$$

where $w_{request}^m$ is a weight defined by the user (e.g. $w_{request}^m = N_m$ would assign higher importance to evacuation requests with larger fleet size).

The routing problem aims at finding the evacuation paths for each request that minimize the weighted average evacuation times (J_R), while accounting for the constraints in the transportation network. This can be formally defined as:

Problem 1. (Routing) given the evacuation requests $\mathcal{R}$ and transportation network $(\mathcal{V}_T, \mathcal{L}_T)$, find the optimal evacuation flows f_{ij}^m for each request r_m by solving:

$$\mathbb{P}_R : \quad \min_{f_{ij}^m} J_R(f_{ij}^m) \quad \text{s.t.} \quad (2) - (6)$$

where J_{RP} is a cost function for the routing problem.

Remark: $\mathbb{P}_R$ is feasible under Assumption 1.

D. Energy Constraints

In practice, electric vehicles have limited driving range, which is dependent on their initial state of charge (SoC). To account for this limitation, we derive a pragmatic energy consumption model in this section.

Assumption 4. *The request r_m has a fleet of N_m vehicles, which are assumed to be homogeneous (same energy consumption characteristics, including start energy e_{start}^m and energy storage capacity E_{max}^m).*

The vehicle homogeneity property rarely holds in practice. There are two approaches to deal with this issue. The first consists in splitting the fleet N_m into sub-groups with similar range characteristics (but at the price of increasing problem complexity). The second approach maintains a single group but bases the planning on the vehicle with the least range in the fleet N_m . Since driving range may vary and is subject to uncertainty, this strategy can be interpreted within a robust optimization framework by treating that vehicle as the worst-case scenario. If it has sufficient energy to complete the route, then all others, with higher driving range, will also succeed. Such a worst-case formulation ensures feasibility for the entire fleet. The latter approach is adopted in this work. While Assumption 4 establishes a uniform baseline demand, its conservatism is explicitly quantified in Appendix E.

The energy consumption of the vehicle fleet N_m is modeled by the following speed-dependent map:

$$g_m(s) = \theta_{e0}^m + \theta_{e1}^m s + \theta_{e2}^m s^2 \quad (8)$$

where g_m is the energy consumption per unit of distance (kWh/mile) for the fleet m , which may vary with the vehicle speed s . In what follows, we assume that the energy consumption increases with vehicle speed:

Assumption 5. $\frac{dg_m(s)}{ds} \geq 0, \quad \forall s \geq 0$

Charging stations may be used to help vehicles gain energy along the evacuation journey. We assume that the charging stations are available in a subset of nodes, $\mathcal{V}_{CS} \subset \mathcal{V}_T$, and can provide up to E_{max}^m of energy to each vehicle.

$$0 \leq \Delta e_j^m \leq E_{max}^m y_j^m \quad (9)$$

$$y_j^m \in \{0, 1\} \quad j \in \mathcal{V}_T \quad (10)$$

where y_j^m is a binary variable that indicates if the vehicle fleet m stops in the charging station j for re-charging. The variable Δe_j^m indicates the amount of energy gained during the stop. If the node does not have a charging station, then vehicles can neither stop nor re-charge in these nodes:

$$y_j^m = 0, \Delta e_j^m = 0, \quad j \in \mathcal{V}_T \setminus \mathcal{V}_{CS} \quad (11)$$

Additionally, vehicles are only allowed to stop in charging stations along the evacuation path, which can be encoded as:

$$y_k^m \leq \sum_{i \in \mathcal{V}^+(k)} x_{ik}^m, \quad k \in \mathcal{V}_T \quad (12)$$

where x_{ij} is a binary variable that indicates if the road segment ij is employed during the evacuation of the fleet $m \in \mathcal{M}$. The energy consumption of a vehicle that traverses a road segment ij with distance d_{ij} and speed s_{ij} is defined as: $d_{ij}g_m(s_{ij})$. From Assumption 3 the speed depends on traffic conditions according to $s_{ij} = d_{ij}/t_{ij}(f_{ij})$. The energy available after the trip ij is modeled as:

$$e_j^m = e_i^m - \phi_{ij}^m(f_{ij})\mathbf{1}_{\{f_{ij}^m > 0\}} + \Delta e_j^m, \quad (i, j) \in \mathcal{L}_T \quad (13)$$

where $\phi_{ij}^m = d_{ij}g_m(\frac{d_{ij}}{t_{ij}(f_{ij})})$ [kWh] is the energy expended while traveling segment ij , e_i^m (and e_j^m) is the energy available to the vehicle at the beginning (and end) of the trip segment. The second term captures the energy lost during the trip segment, which is only active if the vehicles in m use the road segment ij . The initial energy available in the vehicle is defined as

$$e_{om}^m = e_{start}^m \geq 0 \quad (14)$$

for all $m \in \mathcal{M}$. For the path to be physically viable, the energy stored in the vehicle after each road segment must be non-negative:

$$e_j^m \geq 0 \quad j \in \mathcal{V}_T \quad (15)$$

Note also that the energy model (13) is defined for all nodes in the transportation network. We set the energy of these nodes to zero if the fleet m does not visit these nodes; and constraint the maximum energy that the vehicle can hold at any give time to E_{max}^m . These constraints can be encoded as:

$$e_j^m \leq E_{max}^m x_{ij}^m, \quad (i, j) \in \mathcal{L}_T \quad (16)$$

This energy consumption model presented in this section is valid under the following assumption:

Assumption 6. *The fleet of vehicles N_m , with $m \in \mathcal{M}$, follows a single evacuation path.*

This assumption holds that, during a high-stress event, emergency management relies on designating and enforcing recommended evacuation paths [34] to minimize public confusion and thus guarantee coordinated traffic flow.

To fulfill this assumption we include the additional constraint:

$$f_{ij}^m = \alpha_m x_{ij}^m, \quad x_{ij} \in \{0, 1\}, \quad (i, j) \in \mathcal{L}_T \quad (17)$$

Assumption 7. *Each evacuation route may pass through the same node at most once.*

This assumption ensures that the evacuation routes are loop-free, meaning vehicles cannot revisit a previously traveled location. Loop-free routing is a physically realistic requirement critical for both efficiency and safety in an emergency, as revisiting nodes would represent an irrational, time-wasting detour during evacuation. This constraint can be enforced by:

$$\sum_{i \in \mathcal{V}^+(k)} x_{ik}^m \leq 1, \quad k \in \mathcal{V}_T \quad (18)$$

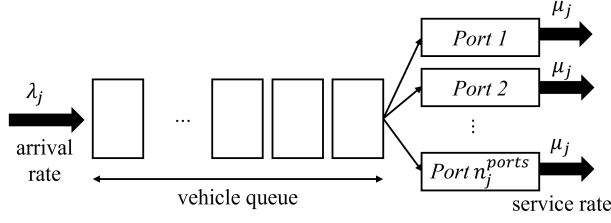


FIGURE 3: Depiction of the EV charging station j , characterized by a vehicle arrival rate of λ_j [car/h], and n_j^{ports} parallel charging ports with a service rate of μ_j [cars/h].

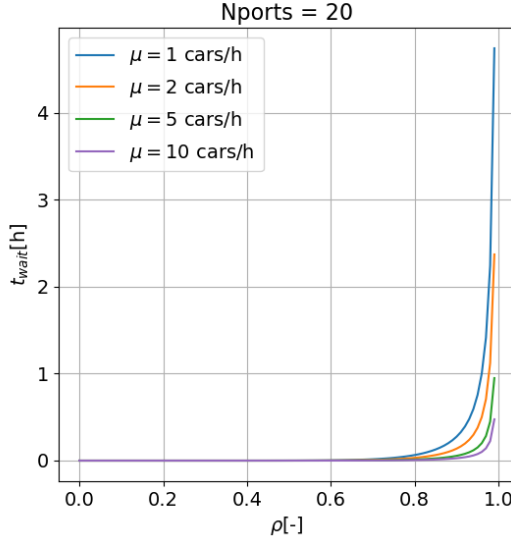


FIGURE 4: Representation of the average waiting time in the queue of the EV charger (t_{wait}) with $c = n^{ports} = 20$ ports for different service rates (μ) and overall utilization ratio ($\rho = \lambda/(c\mu)$).

E. Stop time and Queuing for Re-charging

The overall evacuation time must account for the stoppage time for recharge. This time depends on the number of stations visited during the evacuation, (possible) wait time in the queue to access the charging station, the charging speed and energy transferred to the vehicle for the fleet m . The overall stoppage time can be computed as:

$$T_{stop}^m = \sum_{j \in \mathcal{V}_{CS}} t_{wait,j} y_j^m + t_{setup,j} y_j^m + \frac{1}{p_{ch,j}} \Delta e_j^m \quad (19)$$

where $t_{wait,j}$ is the wait time in the EV charger j , $t_{setup,j}$ is the setup time needed to access the charger (e.g. payment, plug-in the charger, etc.) and $p_{ch,j}$ is the charging power (in kW).

To estimate the wait time in the EV charger we make use of the M/M/c modeling framework from queuing theory [35]. Utilizing an M/M/c queuing structure (and its associated Erlang-C formulas) to model waiting times and server

utilization at electric vehicle charging stations is a well-established and accepted foundation in the transportation and power systems literature (e.g., [12], [16], [36]). This model assumes that vehicles arrive to the charging station j according to a Poisson process with rate λ_j [in cars/h] and are served by $c_j = n_j^{ports}$ identical ports in parallel, each providing service times that are exponentially distributed with rate μ_j [in car/h]. As depicted in Figure 3, each arriving vehicle either finds an available port to charge and is immediately served or it must wait in a queue if all chargers are occupied. The arrival rate of vehicles to the charging station is defined as:

$$\lambda_j = \sum_{m \in \mathcal{M}} \alpha_m y_j^m \quad (20)$$

for all $j \in \mathcal{V}_{CS}$. The service rate $\mu_j = 1/t_{ch,j}$ is dependent on the average charging time for the vehicles $t_{ch,j}$, which can be approximated by:

$$t_{ch,j} = \frac{1}{\mu_j} = t_{setup,j} + \sum_{m \in \mathcal{M}} \frac{\alpha_m}{\sum_{m \in \mathcal{M}} \alpha_m y_j^m} \frac{\Delta e_j^m}{p_{ch,j}} \quad (21)$$

where the second term computes a weighted average of the charger time among all the vehicles that stop in $j \in \mathcal{V}_{CS}$. We use the flow of vehicles in each group (α_m) to weight the relative importance of the charging time of the fleet m . The average waiting time in the queue can be computed using the Erlang C formula [35]:

$$t_{wait,j}(\rho_j, \mu_j) = \frac{r_j^{c_j}}{c_j! c_j \mu_j (1 - \rho_j)^2} p_0(\rho_j, \mu_j) \quad (22)$$

$$p_{0,j}(\rho_j, \mu_j) = \left(\frac{r_j^{c_j}}{c_j! (1 - \rho_j)} + \sum_{n=0}^{c_j-1} \frac{r_j^n}{n!} \right)^{-1} \quad (23)$$

$$\rho_j = \lambda_j / (c_j \mu_j) \quad (24)$$

where $c_j = n_j^{ports}$, $r_j = \lambda_j / \mu_j$ and ρ_j is the overall utilization ratio of the charging station, and $p_{0,j}$ the probability of having zero vehicles in the charging station. As depicted in Figure 4 the average wait time in the queue is close to zero when the utilization ratio is low ($\rho \rightarrow 0$) and increases significantly when $\rho_j \rightarrow 1$. To prevent indefinite queue growth, the overall arrival rate of vehicles must be less than the total service capacity of the charging station (or equivalently $\rho_j \leq 1$), leading to the following stability constraint:

$$\lambda_j \leq n_j^{ports} \mu_j \quad (25)$$

Problem 2. (Routing with Energy Constraints) given the evacuation requests $\mathcal{R}$, transportation network $(\mathcal{V}_T, \mathcal{L}_T)$ and charging network $\mathcal{V}_{CS}$ find the optimal evacuation path x_{ij}^m , stopping charger (y_j^m) and charging energy Δe_j^m for each request r_m by solving:

$$\mathbb{P}_{R+E} : \min_{x_{ij}^m, y_j^m, \Delta e_j^m} J_{R+E}(\cdot), \quad \text{s.t.} \quad (2) - (6), (8) - (25)$$

where J_{R+E} is the driving + stoppage time of the group.

In this work, we consider the average drive + stop time among all evacuation requests:

$$J_{R+E} = \frac{1}{M} \sum_{m \in \mathcal{M}} w_m (T_{drive}^m + T_{stop}^m) \quad (26)$$

F. Mobile Chargers

During emergency evacuations, some of the existing charging infrastructure may not be available (e.g. due to blackouts). In other cases, evacuees may need to take longer detour to access existing charging stations, increasing driving time. To support the evacuations, we assume that there is a list of candidate locations $\mathcal{V}_{MCS} \subset \mathcal{V}_T$ for possible deployment of mobile chargers. In practice, these candidate locations $\mathcal{V}_{MCS}$ should have access to a parking spaces, e.g. a supermarket, a mall, a stadium. In California, many of the counties have emergency plans that identify these locations [37].

In what follows, we assume that the set of available chargers is captured by:

$$\mathcal{V}_{CS} = \mathcal{V}_{FCS} \cup \mathcal{V}_{MCS} \quad (27)$$

where $\mathcal{V}_{FCS}$ is the list of available fixed charging stations. This work focuses on complementing the existing charging infrastructure. Note that deploying MCS to increase FCS capacity can be viewed as effectively expanding existing FCS capabilities, which lies beyond the scope of this paper. As a result:

Assumption 8. $\mathcal{V}_{FCS} \cap \mathcal{V}_{MCS} = \emptyset$

To model the deployment of mobile chargers, let us consider the binary variable $z_i \in \{0, 1\}, i \in \mathcal{V}_{MCS}$ that indicates if the mobile charger i is employed in the evacuation. This indicator should be 1 if the vehicle fleet m stops in the mobile charger, which can be encoded as:

$$y_i^m \leq z_i \quad z_i \in \{0, 1\}, \quad i \in \mathcal{V}_{MCS} \quad (28)$$

Finally, the total number of mobile chargers that can be deployed is fixed:

$$\sum_{i \in \mathcal{V}_{MCS}} z_i = n_{MCS} \quad (29)$$

where n_{MCS} is the total number of mobile charging stations that can be deployed.

Problem 3. (Routing with Energy Constraints and Mobile Chargers Placement) given the evacuation requests $\mathcal{R}$, transportation network $(\mathcal{V}_T, \mathcal{L}_T)$, fixed charging network $\mathcal{V}_{FCS}$, candidate locations for mobile chargers $\mathcal{V}_{MCS}$, find the optimal evacuation path x_{ij}^m , charging energy Δe_j^m , stopping charger (y_j^m) and placement of mobile chargers (z_i) by solving:

$$\begin{aligned} \mathbb{P}_{R+E+M} : \quad & \min_{x_{ij}^m, y_j^m, \Delta e_j^m, z_i} \theta J_{R+E}(\cdot) + (1 - \theta) J_{MCS}(\cdot) \\ \text{s.t.} \quad & (2) - (6), (8) - (25), (27) - (29) \end{aligned}$$

where J_{R+E} is the drive + stoppage time of the group and J_{MCS} is a cost term that capture the cost of deploying mobile chargers and $\theta \in [0, 1]$ is a trade-off factor.

The deployment cost of the mobile charger is captured as:

$$J_{MCS} = \sum_{i \in \mathcal{V}_{MCS}} c_i z_i \quad (30)$$

where c_i is the economic cost of deploying mobile charger i .

III. Evacuation Planning

This section provides three possible approaches to tackle the evacuation problems outlined in the previous section. Figure 5 gives an overview of these approaches.

A. Uncoordinated Evacuation Planning (UEP)

The first approach assumes that each fleet of vehicles m is "selfish" and aims to find the quickest path to reach the destination. We further assume "information blackout", and each fleet m does not know what the other fleets are doing; no mobile chargers are considered. The evacuation plan for each fleet $m^* \in \mathcal{M}$ is computed via a modified version of $\mathbb{P}_{R+E}$:

$$\mathbb{P}^{UEP}(m^*) : \quad \min_{x_{ij}^m, y_j^m, \Delta e_j^m} J_{R+E}(\cdot) \quad (31)$$

$$\text{s.t.} \quad (2) - (6), (8) - (25)$$

$$\alpha_m = 0, \forall m \in \mathcal{M} \setminus \{m^*\} \quad (32)$$

for all $m^* \in \mathcal{M}$. In other words, we compute the optimal routing and charging stop for all the evacuation fleets (one at a time), assuming that no other fleet is present in the transportation/charging network.

Discussion: The UEP is not optimal and may lead to traffic congestion, charging congestion and stranded vehicles. In fact, since every group of vehicles m^* focuses only on their own interest, it may lead to excessive concentration of vehicles in some traffic links and chargers. It may also lead to stranded vehicles due to lack of charging availability. Despite these limitations, the UEP can be used as a baseline to perform a "worst-case" analysis of the evacuation, raising awareness for risk of congestion that can occur due to uncoordinated behavior of evacuees. For example, the evacuation plan generated by $\mathbb{P}^{UEP}(m^*)$ may indicate road segments that are at a high risk of congestion (i.e. where the travel time significantly exceeds the free-flow limit). In anticipation of this scenario, local authorities may increase the number of lanes in this link in order to improve road capacity; they may also identify and communicate alternative routes to evacuees as a means to alleviate (expected) traffic in these critical links. A similar analysis can also be done for chargers that will have high demand.

B. Joint Evacuation Planning (JEP)

In this strategy, we assume that all zones are evacuated simultaneously in a single optimization problem. To simplify the problem, we make a series of practical simplifications.

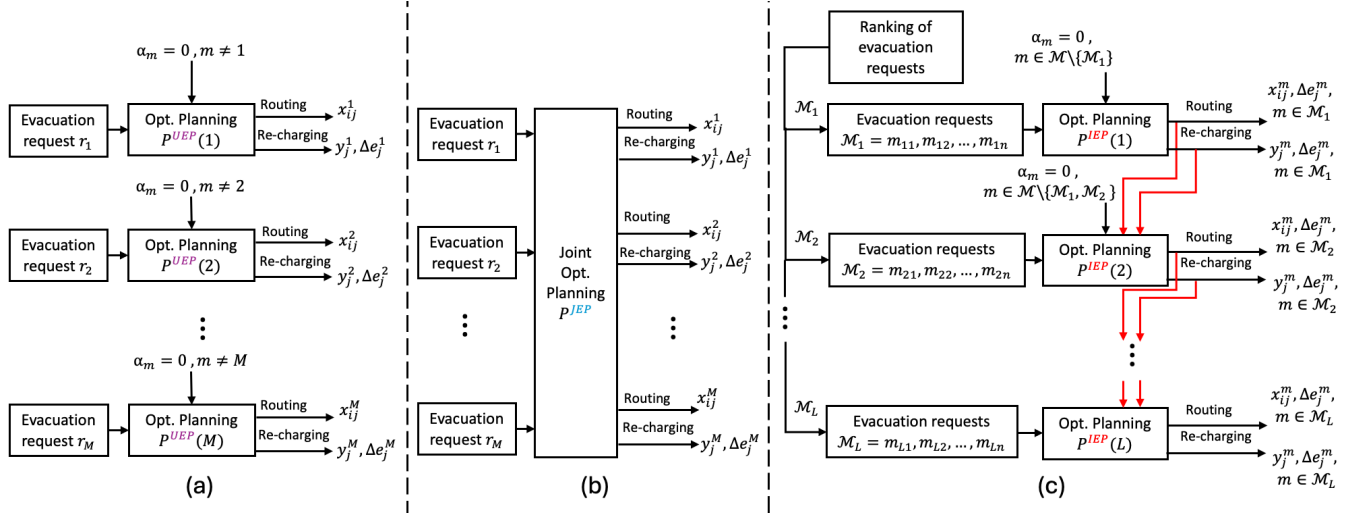


FIGURE 5: Overview of the three evacuation strategies proposed in this work: a) uncoordinated, b) joint, and c) iterative evaluation planning.

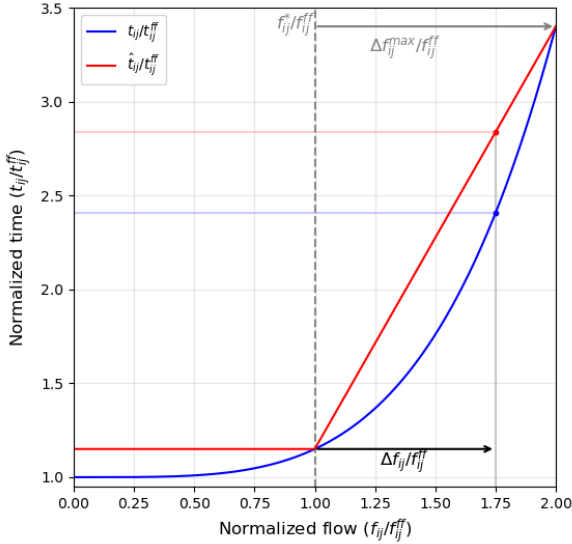


FIGURE 6: Example of the bounding of BPR function t_{ij} with a piecewise-linear approximation $\hat{t}_{ij}$. Note: horizontal and vertical axes normalized with free-flow quantities $(f_{ij}^{ff}, t_{ij}^{ff})$.

Change of variable: since we are only interested in finding a single path for each evacuation request m , we eliminate the variable f_{ij}^m and replace it with $f_{ij}^m = \alpha_m x_{ij}^m$, $x_{ij} \in \{0, 1\}$, which forces all vehicles in the same group to follow the same path.

Driving time: Instead of directly minimizing with respect to the nonlinear BPR function $t_{ij}(f_{ij})$, which was presented in (5), our work considers a linear upper bound. To understand our approach, consider f_{ij}^* as a user-defined maximum

flow limit for the segment (i, j) . This can be set for example $f_{ij}^* = f_{ij}^{ff}$. The vehicle flow is upper bounded by:

$$f_{ij} \leq f_{ij}^* + \Delta f_{ij}, \quad \Delta f_{ij} \geq 0 \quad (33)$$

where Δf_{ij} is a non-negative "slack variable" that indicates the degree of "violation" of the maximum flow constraint. Ideally, we would like to keep $\Delta f_{ij} \rightarrow 0$ to prevent traffic congestion.

Proposition 1. Consider the BPR function $t_{ij}(f_{ij})$ defined in (5) with domain $f_{ij} \in [0, f_{ij}^* + \Delta f_{ij}^{max}] = \mathcal{D}_f$ where $\Delta f_{ij}^{max} \geq 0$ is a user-defined parameter. This BPR function can be upper bounded by $t_{ij}(f_{ij}) \leq \hat{t}_{ij}(\Delta f_{ij})$ for all $f_{ij} \in \mathcal{D}_f$, where

$$\hat{t}_{ij}(\Delta f_{ij}) = \begin{cases} \hat{t}_{ij}^{ff} & \Delta f_{ij} = 0 \\ \hat{t}_{ij}^{ff} + k_{ij} \Delta f_{ij} & \Delta f_{ij} > 0 \end{cases} \quad (34)$$

and Δf_{ij} fulfills (33). The parameters $(\hat{t}_{ij}^{ff}, k_{ij})$ are set as:

$$\hat{t}_{ij}^{ff} = t_{ij}(f_{ij}^*), \quad k_{ij} = \frac{t_{ij}(f_{ij}^* + \Delta f_{ij}^{max}) - t_{ij}(f_{ij}^*)}{\Delta f_{ij}^{max}} > 0 \quad (35)$$

The detailed derivation of the upper-bound formulation is provided in Appendix A. Figure 6 shows this (piece-wise linear) approximation for the BPR function.

This formulation allow us to upper bound the driving time as:

$$T_{drive}^m = \sum_{(i,j) \in \mathcal{L}_T} t_{ij}(f_{ij}) x_{ij}^m \quad (36)$$

$$\leq \sum_{(i,j) \in \mathcal{L}_T} (\hat{t}_{ij}^{ff} + k_{ij} \Delta f_{ij}) x_{ij}^m \quad (37)$$

$$\leq \sum_{(i,j) \in \mathcal{L}_T} \hat{t}_{ij}^{ff} x_{ij}^m + k_{ij} \Delta f_{ij} \quad (38)$$

where the last inequality makes use of the fact that $\Delta f_{ij} x_{ij}^m \leq \Delta f_{ij}$. This cost function provides several advan-

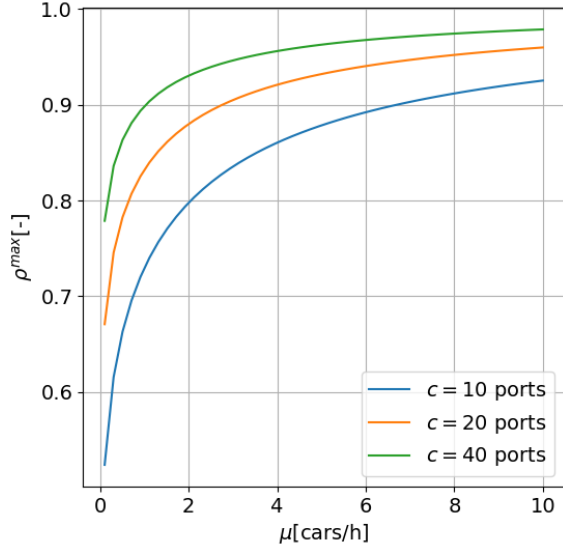


FIGURE 7: Maximum utilization ratio of the EV charging station that guarantees $t_{wait} \leq t_{queue}^{max} = 0.1[h]$. Note: μ is the service rate of the EV charging port.

tages: i) the decision variables $(x_{ij}^m, \Delta f_{ij})$ appear linearly; ii) the designer can explicitly penalize traffic congestion by increasing the value of the parameter k_{ij} ; iii) we also found (empirical) evidence that this alternative formulation was numerically more stable than the original non-linear cost function (7).

Energy Model Simplification The energy model of the vehicle is dependent on the speed, which is affected by traffic congestion and vehicle flow (f_{ij}) along the road segments.

Lemma 1. *The energy consumption is upper bounded by:*

$$\phi_{ij}^m(f_{ij}) \leq \eta_{ij}^m = d_{ij}g_m(s_{ij}^{ff}), \quad \forall f_{ij} \geq 0 \quad (39)$$

where $s_{ij}^{ff} = d_{ij}/t_{ij}^{ff}$ is the free-flow speed and $\eta_{ij}^m \geq 0$ is the upper bound for the energy consumption.

Proof:

According to the traffic model considered in Section A, $s_{ij}^{ff} = d_{ij}/t_{ij}^{ff}$ is the fastest travel time in the road segment ij (obtained under free-flow conditions). Hence $s_{ij}(f_{ij}) \leq s_{ij}^{ff}$ for all $f_{ij} \geq 0$. According to Assumption 5, $g_m(s)$ is non-decreasing, thus $\phi_{ij}^m(f_{ij}) = d_{ij}g_m(s_{ij}) \leq d_{ij}g_m(s_{ij}^{ff}) = \eta_{ij}^m$. ■

This allows us to simplify the energy model with the following linear equality:

$$e_j^m \approx e_i^m - \eta_{ij}^m x_{ij}^m + \Delta e_j^m, \quad \forall ij \in \mathcal{L}_T \quad (40)$$

which provides with a worst-case consumption model for the fleet m .

Simplification of Queuing in EV charging stations

As discussed in Section E, the waiting time in the queue for charging is a nonlinear function of the income/outgoing flow of vehicles to the charging stations. To simplify the formulation of the optimization problem, we will limit the average waiting time that the evacuee may need to wait in the (possible) queue:

$$t_{wait,j}(\rho_j, \mu_j) \leq t_{queue}^{max} \quad (41)$$

where t_{queue}^{max} is a user-defined value for the maximum waiting time in the queue. One can invert the function $t_{wait,j}(\cdot)$ to determine the maximum utilization ratio that fulfills (41):

$$\rho_j^{max}(\mu_j) = t_{wait,j}^{-1}(\mu_j, c_j, t_{queue}^{max}) \quad (42)$$

for $j \in \mathcal{V}_{CS}$. Since $t_{wait,j}(\cdot)$ is monotonically increasing (see Figure 4), the inverse function has a unique solution. Figure 7 depicts $\rho_j^{max}(\mu_j)$ for different values of ports and a $t_{queue}^{max} = 0.1$ [h]. Our idea is to enforce the constraint(41) by limiting the utilization ratio, i.e. $\rho_j \leq \rho_j^{max}(\mu_j)$, which can be re-arranged as

$$\lambda_j \leq \rho_j^{max}(\mu_j) c_j \mu_j \quad (43)$$

where $c_j = n_j^{ports}$.

During an emergency evacuation, we assume that the charging operator limits the maximum charging time ($t_{ch,max}$). This rationing of charging time provides two benefits: i) it prevents formation of large queues in the charger and; ii) help us derive tractable models to enforce (43).

Assumption 9. *i) Charging station $j \in \mathcal{V}_{CS}$ provides at most $E_{ch,j}^{max} = p_{ch,j} t_{ch,max} \geq \Delta e_j^m$ energy to any vehicle; and ii) $t_{setup,j} \ll t_{ch,max}$.*

This assumption implies that charging operators can enforce SoC and time limits during the evacuation. It reflects an operator-managed control strategy rather than unmanaged driver behavior. Commercial fast-charging networks already implement similar protocols during high-demand scenarios; for instance, Tesla's official Supercharger policy [38] imposes automated Congestion Fees once a vehicle reaches 80% State of Charge (SoC) to disincentivize high-SoC dwell delays, while Electrify America [39] enforces automated 85% SoC cutoffs at select locations where drivers receive a 10-minute grace period to move their vehicles before incurring idle fees. Neglecting setup time (e.g., plug-in) is reasonable since it is small compared to charging duration. These assumptions allow a clear bound on service rate for tractable queue modeling. Under these assumptions, the worst-case charging time is given by

$$t_{ch,max} \geq t_{ch,j} \quad \forall j \in \mathcal{V}_{CS} \quad (44)$$

which induces a minimum service rate of $\mu_{min,j} = 1/t_{ch,j}^{max} \leq \mu_j$.

Lemma 2. *Assume that the following inequality holds:*

$$\lambda_j \leq \rho_j^{max}(\mu_{min,j}) c_j \mu_{min,j} \quad (45)$$

Then (43) is satisfied for any $\mu_j \geq \mu_{min,j}$.

Proof:

This result build upon the fact that $\rho_j^{max}(\mu_j)c_j\mu_j$ is an increasing function in μ_j . As a result $\lambda_i\rho_j^{max}(\mu_{min,j})c_j\mu_{min,j} \leq \rho_j^{max}(\mu_j)c_j\mu_j$ for any $\mu_j \geq \mu_{min,j}$, and (45) is fulfilled. ■

The previous result is in accordance with our expectations: if a charging station is able to fulfill the maximum queue time t_{queue}^{max} when the users are using the chargers at "maximum capacity" (i.e. all users maximize their stay in the charger), then t_{queue}^{max} will still be honored when the service rate increases (and the users do not stay the maximum allowed time in the charger). The constraint (45) can be expressed as:

$$\sum_{m \in \mathcal{M}} \alpha_m y_j^m \leq \rho_j^{max} \left(\frac{1}{t_{ch,max}} \right) \frac{n_j^{ports}}{t_{ch,max}} + \Delta\lambda_j \quad (46)$$

for all $j \in \mathcal{V}_{CS}$. The variable $\Delta\lambda_j \geq 0$ is a "slack variable" that allow us to relax the inequality (45). This slack variable will be penalized in the cost function to prevent violation of the maximum queue time.

Remark: if the inequality (46) is satisfied and $\rho_j^{max} \left(\frac{1}{t_{ch,max}} \right) \leq 1$ then the stability condition (25) is also fulfilled.

To further simplify the optimization problem, we use the following approximation for the wait time of the vehicles in fleet $m \in \mathcal{M}$ for all the chargers used along the evacuation:

$$\sum_{j \in \mathcal{V}_{CS}} t_{wait,j} y_j^m \leq \sum_{j \in \mathcal{V}_{CS}} t_{queue}^{max} y_j^m + w_j \Delta\lambda_j^m \quad (47)$$

The first term on the right side is the worst-case waiting time in the queue (t_{queue}^{max}), while the second approximates the increase in waiting time due to the potential violation of (45). The weight w_j can be computed via perturbation analysis:

$$w_j = \frac{d}{d\Delta\lambda} \tilde{t}_{wait,j}(\Delta\lambda) \quad (48)$$

where $\tilde{t}_{wait,j}(\Delta\lambda) = t_{wait,j}(\rho_j^{max}(\mu_{min,j}) + \Delta\lambda, \mu_{min,j})$. Finally, the variable $\Delta\lambda_j^m$ is defined as $\Delta\lambda_j^m = \Delta\lambda_j y_j^m$. This variable captures the increase in input charger flow if the fleet m stops in the charger j (and zero otherwise). To prevent the multiplication of two decision variables, we encode this behavior using the following "big-M" formulation [40]:

$$\Delta\lambda_j^m \geq 0 \quad (49)$$

$$\Delta\lambda_j - M_{big}(1 - y_j^m) \leq \Delta\lambda_j^m \quad (50)$$

$$\Delta\lambda_j^m \leq \Delta\lambda_j + M_{big}(1 - y_j^m) \quad (51)$$

where M_{big} is a large value. If $y_j^m = 0$, then the equality is lifted and $\Delta\lambda_j^m \rightarrow 0$ (as long as $\Delta\lambda_j^m$ is explicitly penalized in the cost function. On the other hand, when $y_j^m = 1$ the above inequalities force $\Delta\lambda_j^m = \Delta\lambda_j$

Final Optimization Problem: Leveraging these approximations, the optimization problem $\mathbb{P}_{R+E+M}$ can be posed as:

$$\begin{aligned} \mathbb{P}^{JEP} : \min & \frac{\theta}{M} \sum_{m \in \mathcal{M}} \left(\sum_{(i,j) \in \mathcal{L}_T} \hat{t}_{ij}^{ff} x_{ij}^m + k_{ij} \Delta f_{ij} \right. \\ & + \sum_{j \in \mathcal{V}_{CS}} t_{queue}^{max} y_j^m + w_j \Delta\lambda_j^m \\ & + t_{setup,j} y_j^m + \frac{1}{p_{ch,j}} \Delta e_j^m \Big) \\ & + (1 - \theta) \sum_{i \in \mathcal{V}_{MCS}} c_i z_i \end{aligned} \quad (52)$$

s.t. $m \in \mathcal{M}$

$$\sum_{i \in \mathcal{V}^+(k)} x_{ik}^m = \sum_{j \in \mathcal{V}^-(k)} x_{kj}^m + \mathbf{1}_{k=d_m} - \mathbf{1}_{k=o_m}, \quad k \in \mathcal{V}_T \quad (53)$$

$$x_{ij}^m \in \{0, 1\}, \quad (i, j) \in \mathcal{L}_T \quad (54)$$

$$\sum_{i \in \mathcal{V}^+(k)} x_{ik}^m \leq 1, \quad k \in \mathcal{V}_T \quad (55)$$

$$f_{ij} = \beta_{ij} + \sum_{m \in \mathcal{M}} \alpha_m x_{ij}^m \leq f_{ij}^* + \Delta f_{ij},$$

$$\Delta f_{ij} \geq 0 \quad (56)$$

$$e_{o_m}^m = e_{start}^m \geq 0 \quad (57)$$

$$e_j^m = e_i^m - \eta_{ij}^m x_{ij}^m + \Delta e_j^m, \quad (i, j) \in \mathcal{L}_T \quad (58)$$

$$e_j^m \geq 0 \quad j \in \mathcal{V}_T \quad (59)$$

$$e_j^m \leq E_{max}^m x_{ij}^m, \quad (i, j) \in \mathcal{L}_T \quad (60)$$

$$0 \leq \Delta e_j^m \leq E_{max}^m y_j^m \quad j \in \mathcal{V}_T \quad (61)$$

$$y_j^m \in \{0, 1\}, \quad j \in \mathcal{V}_T \quad (62)$$

$$y_j^m = 0, \Delta e_j^m = 0, \quad j \in \mathcal{V}_T \setminus \mathcal{V}_{CS} \quad (63)$$

$$y_k^m \leq \sum_{i \in \mathcal{V}^+(k)} x_{ik}^m, \quad k \in \mathcal{V}_T \quad (64)$$

$$\sum_{m \in \mathcal{M}} \alpha_m y_j^m \leq \rho_j^{max} \left(\frac{1}{t_{ch,max}} \right) \frac{n_j^{ports}}{t_{ch,max}} + \Delta\lambda_j, \quad (65)$$

$$\Delta\lambda_j^m \geq 0 \quad (66)$$

$$\Delta\lambda_j - M_{big}(1 - y_j^m) \leq \Delta\lambda_j^m \leq \Delta\lambda_j + M_{big}(1 - y_j^m)$$

$$y_i^m \leq z_i, \quad i \in \mathcal{V}_{MCS} \quad (67)$$

$$\sum_{i \in \mathcal{V}_{MCS}} z_i = n_{MCS} \quad (68)$$

$$z_i \in \{0, 1\} \quad i \in \mathcal{V}_{MCS} \quad (69)$$

The boolean decision variables are x_{ij}^m, y_i^m, z_i ; The non-negative integer decision variables are $\Delta e_j^m, e_j^m, f_{ij}, \Delta f_{ij}, \Delta\lambda_j, \Delta\lambda_j^m$;

In what follows, we will use χ to denote the optimal solution of $\mathbb{P}^{JEP}$ (or its variants).

C. Iterative Evacuation Planning (IEP)

In many cases, the evacuation occurs in multiple phases. Neighborhoods that are closer to danger are usually asked to leave earlier. Areas that are further away have more time to leave; and may wait to see if the natural hazard (e.g. wildfire) changes course or vanishes.

To understand the principle of operation of the IEP, let us assume that the evacuation requests are ranked based on proximity to danger. Let $M_{\text{danger}} = \{M_1, M_2, \dots, M_L\}$ be a collection of disjoint subsets of M , where each subset $M_l = (m_{l1}, m_{l2}, \dots, m_{ln})$ contains a total of M_l evacuation requests with the same level of proximity to danger. The subsets are ordered such that any origin in M_l is closer to the danger zone than any origin in M_{l+1} . In other words, the origins $o_{m_{l1}}, o_{m_{l2}}, \dots, o_{m_{ln}}$ are closer to the danger than the origins $o_{m_{(l+1)1}}, o_{m_{(l+1)2}}, \dots, o_{m_{(l+1)n}}$.

We solve the IEP one zone at a time. At each planning iteration k we assume that i) vehicle fleet k has knowledge of the evacuation plans of the fleets that were previously evacuated ($k-1, k-2, \dots, 1$); ii) future evacuation fleets are not considered in the iteration k :

$$\alpha_{m \in M_l} = 0, \quad l = k+1, \dots, L \quad (70)$$

The overall IEP planning can be computed by recursively solving a modified version of the JEP problem. For each $k^* = 1, \dots, L$ we solve:

$$\begin{aligned} \mathbb{P}^{IEP}(k^*) : \min & \frac{\theta}{\sum_{l=k^*}^L M_l} \sum_{m \in M_l, l=1, \dots, k^*} \left(\sum_{(i,j) \in \mathcal{L}_T} \hat{t}_{ij}^{ff} x_{ij}^m \right. \\ & + k_{ij} \Delta f_{ij} + \sum_{j \in \mathcal{V}_{CS}} t_{queue}^{max} y_j^m + w_j \Delta \lambda_j^m \\ & \left. + t_{setup,j} y_j^m + \frac{1}{p_{ch,j}} \Delta e_j^m \right) \\ & + (1 - \theta) \sum_{i \in \mathcal{V}_{MCS}} c_i z_i \end{aligned} \quad (71)$$

$$\text{s.t. } (53) - (69), \quad \mathcal{M} = \mathcal{M}_{k^*} \quad (72)$$

$$\alpha_{m \in M_l} = 0, \quad l = k^* + 1, \dots, L \quad (73)$$

$$\begin{aligned} x_{ij}^m &= \tilde{x}_{ij}^m, \quad y_j^m = \tilde{y}_j^m \\ f_{ij}^m &= \tilde{f}_{ij}^m, \quad \Delta e_j^m = \Delta \tilde{e}_j^m \\ (i,j) &\in \mathcal{L}_T \\ m \in M_l, \quad l &= 1, \dots, k^* - 1 \end{aligned} \quad (74)$$

$$\begin{aligned} z_i &\geq \tilde{z}_i^l, \\ i \in \mathcal{V}_{MCS}, \quad l &= 1, \dots, k^* - 1 \end{aligned} \quad (75)$$

where $\tilde{x}_{ij}^{m_k}, \tilde{y}_j^{m_k}, \tilde{f}_{ij}^{m_k}, \Delta \tilde{e}_j^{m_k}, \tilde{z}_i^{m_k}$ are the solutions found in the previous iterations of the IEP planning ($k = 1, \dots, k^* - 1$) and are assumed to be fixed during the iteration k^* . The constraint (75) forces the mobile chargers deployed in the previous iterations to remain fixed in their positions (no redeployment allowed).

From a practical perspective, the IEP gives higher priority to vehicle fleets that are closer to danger; the price to pay is

a (potential) increase in evacuation time for the zones away from danger, which receive less priority in the access to the transportation/charging network.

One may wonder if this prioritization is fair. The IEP can be re-arranged to take into account other factors. For example, evacuation requests may be ranked based on the number of people that departs in each request or available energy in the vehicle at the beginning of the journey. This would give higher priority to large vehicle fleets or to vehicles with lower SoC. Different emergency managers and communities have different preferences and priorities. The IEP provides a flexible framework that managers can use to encode priority factors in the evacuation planning.

Since the number of decision variables for IEP in each iteration is less than or equal to that of JEP, IEP offers a computational advantage when planning for high-risk groups. However, it typically requires a larger number of iterations to plan the entire evacuation group, and as a result, the overall solution time of IEP can be longer than that of JEP. (see Section 3.G.)

D. Discussion and Analysis

Remark: all costs and constraints are linear in the unknown variables. The problem can be efficiently solved using modern *linear mixed-integer programming* tools, such as CPLEX [41] or Gurobi [42]. However, the branch-and-bound algorithm commonly used for Mixed-Integer Linear Programs (MILPs) may require exponential time to achieve a 0% optimality gap [43]. Such a limitation highlights the potential computational burden of solving large-scale instances.

1) Solution without re-charging

Intuitively, one expects that, if the vehicles have enough energy to reach the destination, then no stop for re-charging will be needed. This section provides a mathematical formulation of this insight. First we introduce a particular solution for the evacuation problem:

Definition 2. The congestion-aware charging-free (CACF) optimal solution is defined as $\tilde{\chi} = (\tilde{x}_{ij}^m, \Delta \tilde{f}_{ij}^m, \tilde{y}_j^m, \Delta \tilde{e}_j^m, \tilde{z}_i)$, where $(\tilde{x}_{ij}^m, \Delta \tilde{f}_{ij}^m)$ is obtained by solving $\mathbb{P}^{JEP}$ - defined in Appendix D and

$$\begin{aligned} \tilde{y}_l^m &= 0, \quad \Delta \tilde{e}_l^m = 0, \quad \tilde{z}_i = 0 \\ m \in \mathcal{M}, (i,j) &\in \mathcal{L}_T, l \in \mathcal{V}_T, i \in \mathcal{V}_{MCS}. \end{aligned} \quad (76)$$

In the above definition $\mathbb{P}^{JEP}$ denotes a simplified problem that focus on finding the shortest path to the destinations, considering traffic congestion but no energy constraints (see Appendix D).

Proposition 2. Assume that the following condition holds for all $m \in \mathcal{M}$

$$e_{start}^m \geq \sum_{(i,j) \in \mathcal{L}_T} \eta_{ij}^m \tilde{x}_{ij}^m \quad (77)$$

Then $\tilde{\chi}$ is a feasible and optimal solution for $\mathbb{P}^{JEP}$.

Detailed derivation of this Proposition is provided in Appendix B.

This result will mostly hold if the evacuation distance is relatively small (e.g. a few miles) and the vehicles can easily reach safety.

2) Feasibility with charge at home before departure

If the initial energy in the vehicles is not enough to reach the destination, evacuees may charge at home to gain enough driving range to reach destination without stopping along the way. The following result provides sufficient conditions for feasibility of this strategy

Proposition 3. Assume that i) all origins of the evacuation request $m \in \mathcal{M}$ have a fixed charger($o_m \in \mathcal{V}_{FCH}$) and ii) the following conditions holds:

$$E_{max}^m \geq \sum_{(i,j) \in \mathcal{L}_T} \eta_{ij}^m \tilde{x}_{ij}^m - e_{start}^m \quad (78)$$

for all $m \in \mathcal{M}$. Then $\mathcal{P}^{JEP}$ is feasible with candidate solution:

$$\begin{aligned} x_{ij}^m &= \tilde{x}_{ij}^m, \quad \Delta f_{ij} = \Delta \tilde{f}_{ij}, \quad (i,j) \in \mathcal{L}_T \\ y_j^m &= 1, \quad j = o_m, \\ \Delta e_j^m &= \max(0, \sum_{(i,j) \in \mathcal{L}_T} \eta_{ij}^m \tilde{x}_{ij}^m - e_{start}^m), j = o_m, \\ y_j^m &= 0, \quad \Delta e_j^m = 0, \quad j \in \mathcal{V}_{CS} \setminus \{o_m\} \\ z_i &= 0, \quad i \in \mathcal{V}_{MCS} \end{aligned} \quad (79)$$

$$\begin{aligned} y_j^m &= 0, \quad \Delta e_j^m = 0, \quad j \in \mathcal{V}_{CS} \setminus \{o_m\} \\ z_i &= 0, \quad i \in \mathcal{V}_{MCS} \end{aligned} \quad (80)$$

for all $m \in \mathcal{M}$.

The full derivation supporting this proposition is given in Appendix C. Note that the above result only guarantees feasibility. The option of "charging at home" may not be optimal. Normally at-home chargers are slower than (fast) public chargers, and may increase the evacuation time.

3) De-Activating Energy Constraints

The energy constraint (58) can be evaluated at any node in the network. In practice, it only makes sense to consider this constraint when the node is used in the evacuation. One way to encode this activation/de-activation of the constraint is via the "big-M" method [40], where (58) is replaced by:

$$e_j^m \geq e_i^m - \eta_{ij}^m x_{ij}^m + \Delta e_j^m - M(1 - x_{ij}^m) \quad (81)$$

$$e_j^m \leq e_i^m - \eta_{ij}^m x_{ij}^m + \Delta e_j^m + M(1 - x_{ij}^m) \quad (82)$$

where M is a very large constant. When $x_{ij}^m = 1$, we obtain (58). This equality becomes "inactive" for $x_{ij}^m = 0$.

IV. Validation Results with Low-Fidelity Models

In this section, we present evacuation charging and routing solutions using three different approaches: UEP, JEP, and IEP.

A. Traffic network

The evacuation planning approaches were implemented on the Sioux Falls map from [15], [44], and [45], as shown in Figure 8. In the traffic network, fixed charging stations (represented as square-shaped nodes) are pre-positioned at nodes 4, 10, 11, 14, and 15, as proposed in [15]. A hazard is assumed to occur 0.3 miles south of node 19, threatening the nearby nodes 19, 18, 20, and 21, which are located 0.30, 1.20, 1.40, and 1.41 miles from the hazard, respectively. Each affected origin node is assigned a constant evacuation demand toward four destination nodes that are farthest from the hazard: nodes 0, 1, 2, and 12.

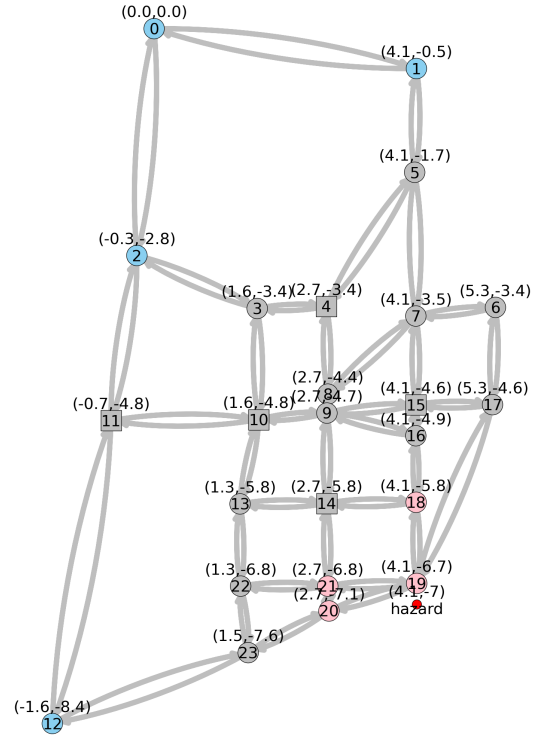


FIGURE 8: Sioux Falls Evacuation map with hazard (red), evacuation origins (pink) and destinations (blue) locations.

B. Low fidelity simulation model for evacuation time calculation

The evacuation planning solutions generated using the proposed planning approaches were fed into a low-fidelity model to compute and compare the total evacuation time. In this model, the average driving time on each link is estimated using the Bureau of Public Roads (BPR) function from (5), with $b_1 = 0.15$ and $b_2 = 4$, which are commonly used in the literature [46], [33]. We assume vehicles at free-flow condition travels with speed $s = 55$ mph, which is commonly used as the speed limit for two-lane undivided

Parameter	Value	Unit	Notes
s^{ff}	55	mph	Average driving speed without congestion.
E_{max}^m	64	kWh	2020 Kia Soul EVs battery size [47].
$p_{ch,j}$	10/50	kW	Output power: Level 2 fixed EV charger [48] / DC fast charging mobile EV charger [49].
$g_m(s^{ff})$	0.26	kWh/mile	energy consumption per unit of distance for the fleet N_m [47].

TABLE 3: List of evacuation time modeling parameters

Parameter	Value	Unit	Notes
k_{ij}	20		Slope for BPR function linear upper bound.
w_j	0.25		The cost of potential violation of worst-case waiting time.
t_{queue}^{max}	0.5	hour	The worst-case waiting time in the queue.
$t_{ch,max}$	0.2	hour	Worst-case charging time.
t_{setup}	0	hour	setup time at charging station.

TABLE 4: List of optimization problem parameters

roadways in the US. The total driving time is calculated from (6).

We assume that vehicles are charged at a constant rate per unit of driving range. All fixed chargers are assumed to be L2 chargers with output power $p_{ch,j} = 10$ kW, and all mobile chargers are assumed to be Direct Current (DC) fast charging with output power $p_{ch,j} = 50$ kW. The EVs are modeled as 2020 Kia Soul EVs with a 64 kWh battery pack and a full driving range of 243 miles [47]. The average energy consumption per unit of distance for the EV is calculated as $g_m = 64/243 = 0.26$ kWh/mile. The wait time at the charging station is calculated from (22). The total evacuation time for each OD pair is then calculated as the sum of driving time, charging time and waiting time.

The parameters used to calculate evacuation time are summarized in Table 3, while those used to formulate the optimization problem are listed in Table 4. Table 5 presents the use cases considered to evaluate the proposed evacuation planning approaches.

C. Use Case 1: Road Traffic Aware Routing

In this section, we compare the evacuation time under heavy traffic flow using different evacuation approaches on the Sioux Falls map. We assume an initially empty network, with each Origin-Destination (OD) pair assigned a flow size of 4000 vehicles per hour. The flow size of 4000 vehicles/hour per OD pair was intentionally chosen as an extreme stress test for the network to clearly illustrate how the different planning approaches behave under severe,

worst-case evacuation demand. The initial state of charge (SoC) of each vehicle was set to 100%, ensuring that no charging was required during evacuation. Figure 9 compares evacuation times for different OD pairs under various evacuation planning approaches, and shows that both JEP and IEP significantly reduce travel times compared to the UEP approach. Figure 11 shows the normalized vehicle flow for the three evacuation strategies. While the flow-size-to-free-flow-capacity ratio ($\frac{f_{ij}}{f_{ij}^{ff}}$) on the traffic network in the UEP approach exceeds 3.45 on the traffic network, both the JEP and IEP approaches maintain significantly lower ratios, with a maximum around 1.23. Figure 10 reveals that IEP prioritizes the evacuation of OD pairs closest to the hazard, originating from node 19, with the cost of increased travel times for less-at-risk OD pairs, nodes 18, 20, 21. To make this fairness trade-off concrete, we evaluate the Worst-Off-Zone Delay Metric ($\Delta T = T_{IEP} - T_{JEP}$), which captures the delay penalty incurred by non-priority zones relative to the JEP social optimum. As shown in Figure 10, Node 18 (the second-closest origin to the hazard) experiences the largest delay penalty in the network, with its average travel time increasing from 0.23 h under JEP to 0.29 h under IEP ($\Delta T = +0.06$ h = +3.6 min, a 26.1% delay penalty). Meanwhile, the furthest origin (Node 21, located 1.41 miles from the hazard) incurs a smaller delay of $\Delta T = +0.04$ h (+2.4 min, a 19.0% increase). This concrete metric demonstrates that IEP imposes a bounded, localized penalty on intermediate corridors to guarantee rapid clearance for origin Node 19 in immediate danger.

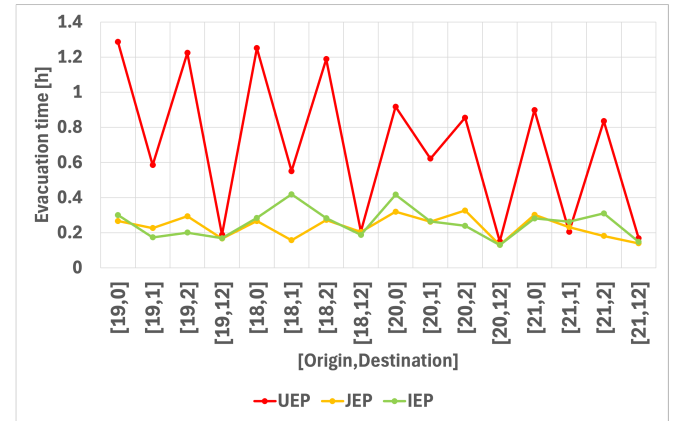


FIGURE 9: Use Case 1 – Evacuation time for UEP, JEP, and IEP. (Origin-destination pairs ordered from left to right by increasing origin-to-hazard distance.)

D. Use Case 2: Charging Traffic Aware Routing

In the second use case, we reduced the initial SoC level of each vehicle to 20% and increase the map scale by a factor of 9 to demonstrate how charging demands affect evacuation time, especially queuing time, in long-distance evacuation scenarios. Out of the 16 OD pairs, 14 require charging to

TABLE 5: Simulation Use Cases and Parameters

#	Use Case	Methods	Num of OD Pairs	Num of ODs needing charge	SoC ₀	Sum Flow per Origin	Flow per OD	FCS Num	FCS Power	Num of FCS Ports	MCS Num	MCS Power	Num of MCS Ports	Map Scale	β
1	Road Traffic Aware Routing	UEP/IEP/JEP	16	0	100% (no stop)	16000 cars/h	4000 cars/h	–	–	–	–	–	–	1x	0
2	Charging Traffic Aware Routing	UEP/IEP/JEP	16	14	20%	20 cars/h	5 cars/h	5	10 kW	20	–	–	–	9x	0
3	Mobile Charger Impact	JEP/IEP	16	14	20%	20 cars/h	5 cars/h	0	–	–	1/2/3	50 kW	10	9x	0
4	Impact of SoC ₀	JEP	16	0 to 14	20%–40%	20 cars/h	5 cars/h	5	10 kW	20	–	–	–	9x	0

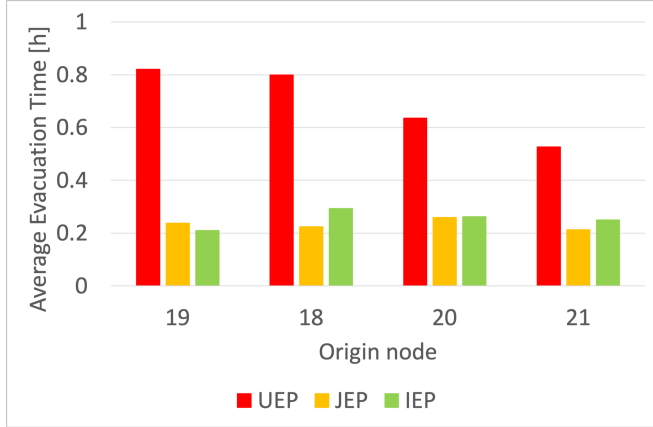


FIGURE 10: Use Case 1 - Average evacuation time for origin-destination pairs with a common origin, ordered from left to right by increasing origin-to-hazard distance.

reach their destinations. The OD pairs [20, 12] and [21, 12] do not require charging. We assume that each fixed charging station has 20 charging ports, and the flow size for each OD pair is set to 5 cars per hour.

Figure 12a summarizes the waiting times at FCSs for each OD pair under this scenario. The UEP approach does not account for waiting times at charging stations, which results in uneven utilization across the network and long queues at FCS 4. Figure 12b shows that under the UEP approach, only 3 out of 5 FCSs are utilized. In particular, FCS 4 experiences an extremely high utilization ratio. With $\rho_4 > 1$, the EV arrival rate exceeds the station's service rate, causing an unstable system where the queue grows without bound. Unlike UEP, the JEP and IEP approaches actively redirect traffic to alternative FCSs to avoid long queues at FCS 4. Figure 13 illustrates this behavior using OD pair [19, 2] as an example.

E. Use Case 3: Mobile Charger Impact

In the third use case, we analyze the use of MCSs to aid evacuations. We adopt the same evacuation zone setup described in Section D, but assume all FCS in the network

become non-operational due to a power grid failure caused by the hazard ($\mathcal{V}_{FCS}^{case3} = \emptyset$). To maintain charging support for ZEVs during evacuation, we assume that MCSs can be rapidly deployed at the same nodes where FCSs were originally located ($\mathcal{V}_{MCS} \subset \mathcal{V}_{FCS}^{case2}$). These nodes are likely to offer pre-existing space, roadway access, and established demand patterns, making them logical fallback deployment sites. This setup enables us to evaluate the feasibility and effectiveness of MCSs as a resilient, flexible alternative when grid-dependent charging infrastructure is compromised. We assume each MCS is equipped with 10 DC fast charging ports, and each port is capable of delivering 50 kW of output power.

Figure 14 summarizes the changes in evacuation time for IEP and JEP as the number of MCS increases from 1 to 3. An increase in the number of MCSs from 1 to 2 results in a 14% reduction in evacuation time under JEP and a 21% reduction under IEP, highlighting the impact of additional charging resources on evacuation efficiency. However, further increasing the number of MCSs from 2 to 3 results in minimal improvement, suggesting diminishing returns. Notably, under the 1-MCS scenario, the JEP approach outperforms IEP by achieving a lower average evacuation time and reduced variability. This performance gap narrows as more MCSs are added, with both approaches converging to similar evacuation times at 2 and 3 MCSs. The error bars show that the variability in evacuation time is highest when only 1 MCS is available, particularly under the IEP approach, due to congestion or imbalanced access to charging infrastructure.

Figure 15 shows the average MCS utilization ratio as a function of the number of MCSs deployed under the JEP and IEP approaches. As expected, the average utilization ratio decreases as more MCSs are added, due to the additional charging capacity being distributed across the network. When only one MCS is deployed, both approaches show high utilization ratios (close to 1), indicating that the charging infrastructure is operating near its maximum capacity, which can lead to long queues and delays. With two MCSs, the utilization drops sharply to approximately 0.5, and further declines to around 0.35 when three MCSs are deployed. This trend reflects the diminishing marginal load on each

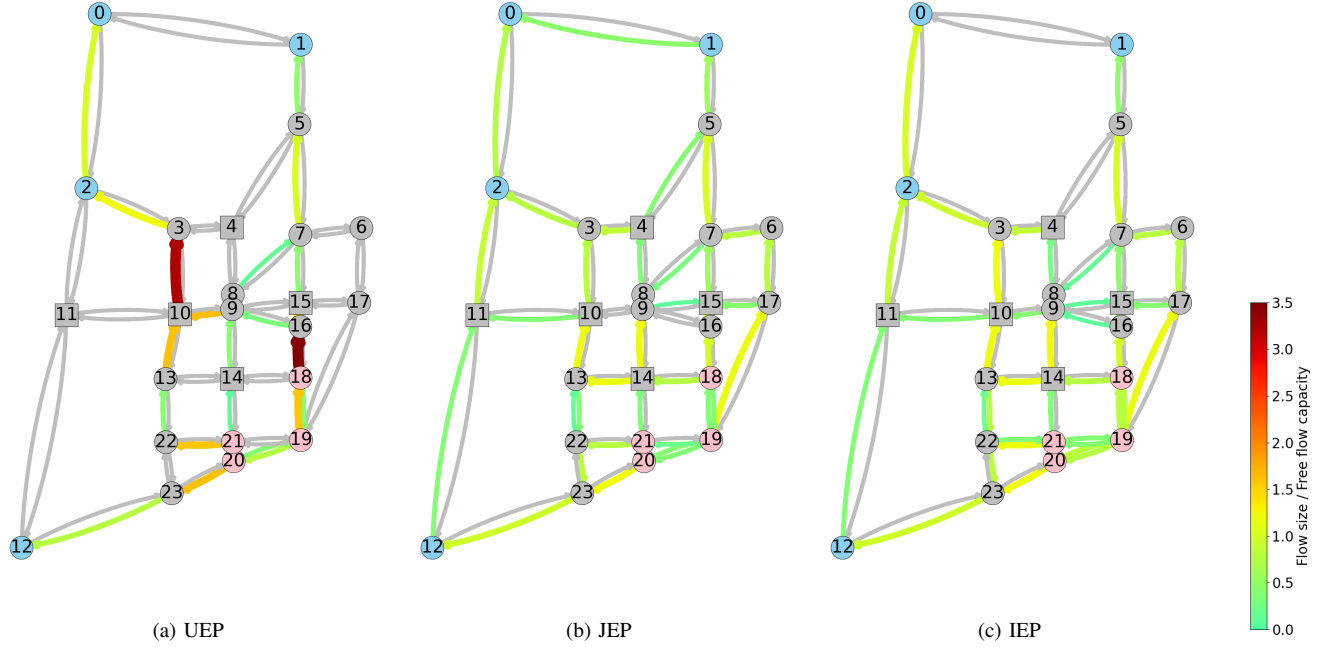


FIGURE 11: Use Case 1 - Normalized vehicle flow size.

station as more MCSs become available. Notably, the JEP and IEP curves closely follow each other, suggesting that both strategies result in similar levels of MCS utilization across this use case. However, the JEP approach consistently maintains a slightly lower utilization ratio, indicating a more balanced allocation of charging demand across available stations due to its coordinated planning.

F. Use Case 4: Initial SoC

In this use case, we analyze the impact of an EV's initial SoC on evacuation performance. We consider the JEP method to plan the evacuation. The evacuation zone setup described in Section D was adopted, with the initial SoC for vehicles, e_{start} , varying from 20% to 40%. Figure 16a summarizes the distribution of evacuation times under varying initial SoC levels, revealing a clear inverse relationship between initial SoC and evacuation time. Figure 16b illustrates the FCSs utilization ratio under different initial SoC, and suggests that the improved evacuation times at higher initial SoC levels result from a reduced need for charging. The additional time spent during evacuation due to charging can be attributed to three primary factors: (i) detours to reach charging stations, (ii) the time required to charge the vehicle, and (iii) waiting time in the charging queue. Figure 17 illustrates these three sources of delay using the example of OD pair [20,1] under different initial SoC levels. Driving time shows a slight improvement from 20% to 25% SoC, suggesting that a shorter route is selected when less charging is needed. As the initial SoC increases from 20% to 30%, charging time decreases significantly, with vehicles starting at 30% SoC

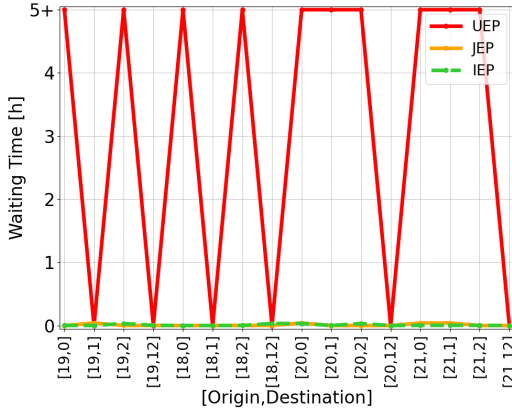
requiring no additional charging. Waiting time remains low and negligible across all SoC levels, indicating minimal queuing delays at charging stations. The total evacuation time is highest at 20% SoC and decreases steadily with increasing SoC, primarily due to reduced charging requirements. These results highlight the importance of maintaining a higher initial SoC to minimize charging delays and enhance overall evacuation efficiency.

V. Validation Results with High-fidelity Model

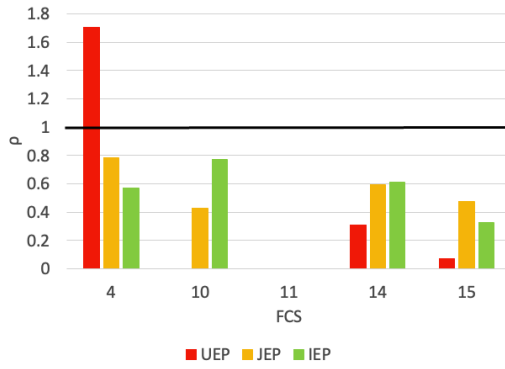
In this section, we evaluate the UEP, JEP, and IEP methods using Simulation of Urban MObility (SUMO) [50], a high-fidelity traffic simulator.

A. Mariposa traffic network

We model an evacuation in Mariposa, CA, caused by a wildfire. Mariposa is frequently subject to a high risk of natural disasters such as wildfires, floods, and landslides. Figure 18 illustrates this vulnerability by mapping recent significant wildfires between 2000 and 2023. This figure demonstrates that the town of Mariposa has been in close proximity to several large-scale incidents, most notably the Detwiler Fire (81,826 acres) to the west and the Oak Fire (19,251 acres) to the north. To analyze evacuation dynamics in this high-risk environment, the simulation utilizes a simplified map of the region, as shown in Figure 19. The network represents a simplified evacuation graph of the Mariposa region, constructed to capture the main corridors and junctions relevant to evacuation routing while abstracting away minor local streets. Nodes denote key intersections,



(a) Waiting time. (Origin-destination pairs ordered from left to right by increasing origin-to-hazard distance.)



(b) FCS utilization ratio

FIGURE 12: Use Case 2 - Waiting time at FCS for each OD pair (a), FCS utilization ratio (b).

population centers, or control points, while edges represent the connecting road segments. At the heart of this system, the upper central cluster (near nodes 14, 15) represents the Mariposa town area, where a dense subnetwork of short links facilitates local movement. Extending outward, the network exhibits a tree-like, radial structure as local branches converge into major corridors. These outlets lead to larger cities via destination nodes 0, 7, 60, and 49, which correspond to State Routes 59, 140, 145, and 41, respectively. The full map implementation is visualized in the SUMO screenshot in Figure 20a. To maintain a controlled environment for the capacity analysis, the model assumes single-lane roads for all edges, and signalized intersections at the nodes shown in Figure 20b.

B. Vehicle model

Consistent with the Sioux Falls evaluation, the 2020 Kia Soul EV model was implemented in SUMO using parameters adapted from [52]. Table 6 lists the detailed parameters used for this EV. Figure 21 shows a visualization of a vehicle in

TABLE 6: Vehicle Parameters adapted from [52]

Category	Parameter	Value
General	Color	White
	Emission Class	Energy/unknown
	Mass	1830 kg
Kinematics	Max Speed	29.06 m/s
	Min Gap	2.50 m
	Acceleration	20.0 m/s ²
	Deceleration	20.0 m/s ²
	Sigma (Driver Imperfection)	0.0
Battery/Engine	Battery Capacity	64,000 Wh
	Maximum Power	150,000 W
	Propulsion Efficiency	0.98
	Recuperation Efficiency	0.96
	Constant Power Intake	100 W
Aerodynamics/Drag	Front Surface Area	2.6 m ²
	Air Drag Coefficient	0.35
	Roll Drag Coefficient	0.01
	Radial Drag Coefficient	0.1
	Rotating Mass	40 kg
Control	Stopping Threshold	0.1 m/s

the SUMO GUI stopped at intersection node 5 while waiting for the traffic light to turn green.

C. Charging station model

In SUMO, an EV charging station is modeled as a *chargingStation* object placed on a lane, where electric vehicles equipped with a battery device can stop and recharge. As illustrated in Figure. 22a, to model a charging station associated with a traffic node (node 44), an auxiliary node (node *CS44*) is introduced 350 m northeast of the original traffic node. The connecting link between the traffic node and the auxiliary charging node is then used to place the charging station. This construction represents charging as a short detour from the main traffic network, allowing vehicles to enter, dwell for charging, and rejoin traffic without interfering with intersection operations. In Figure. 22a, the green marker indicates an actively used charging port, while the red markers denote charging ports that are not in use. Since the *chargingStation* object in SUMO does not have parking areas for vehicles waiting to charge, additional lanes were created on the link to provide additional waiting areas for vehicles waiting in the queue, as shown in Figure 22b and Figure 22c. The charging durations for vehicles are determined based on the required energy demands, assuming 100% charging efficiency. Detailed charging station parameters are defined in Table 7.

D. BPR function calibration

The BPR function is calibrated for the evacuation route from node 5 to node 2 and from node 31 to node 45, as these road segments constitute the primary bottleneck in

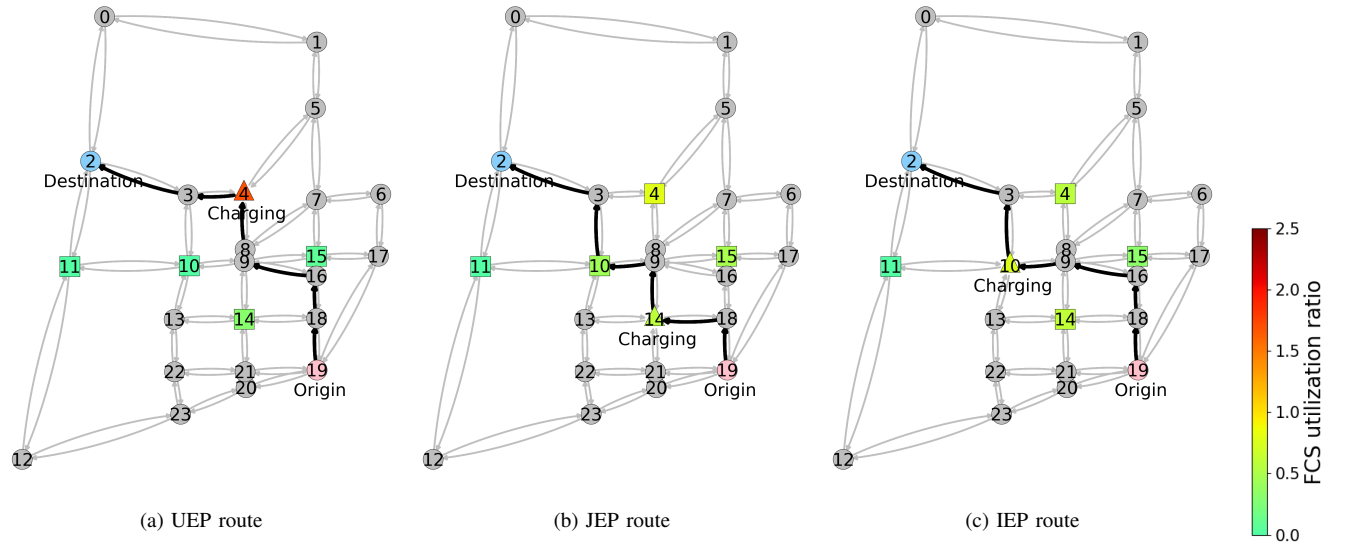


FIGURE 13: Use Case 2 - Routing and charging plan for the OD pair [19,2] with different planning approaches.

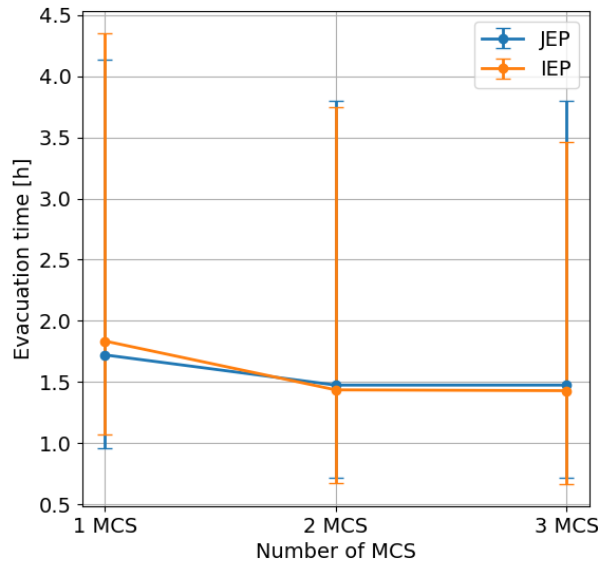


FIGURE 14: Use Case 3 - Evacuation Time vs. Number of MCSs Deployed for JEP and IEP Approaches.

the evacuation scenario and dominate congestion formation. Traffic from multiple upstream branches converges onto these segments, making their performance critical to overall evacuation efficiency. Calibrating the BPR function on these routes ensures that the macroscopic congestion model accurately captures the capacity-limiting behavior observed in the high-fidelity SUMO simulation, thereby providing a reliable representation of travel time growth under increasing evacuation demand.

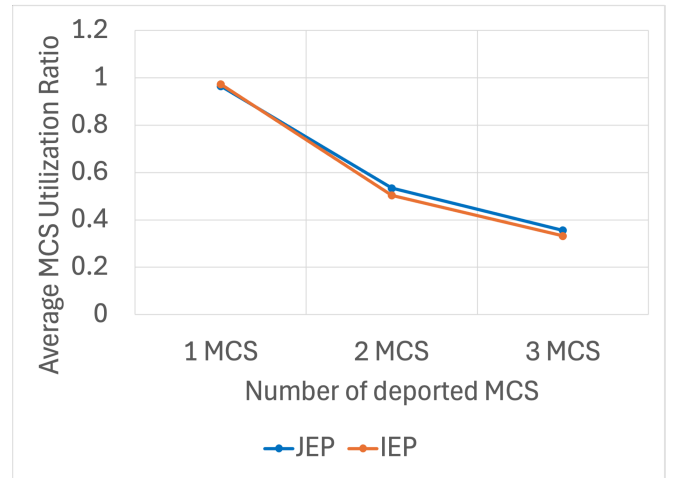
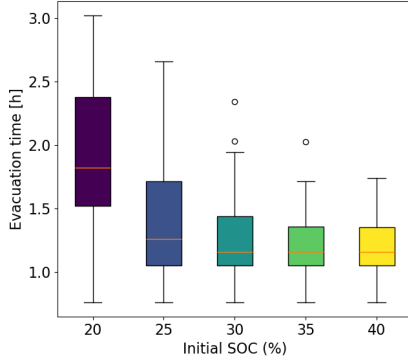


FIGURE 15: Use Case 3 -Average MCS Utilization ratio vs. Number of MCSs Deployed for JEP and IEP Approaches.

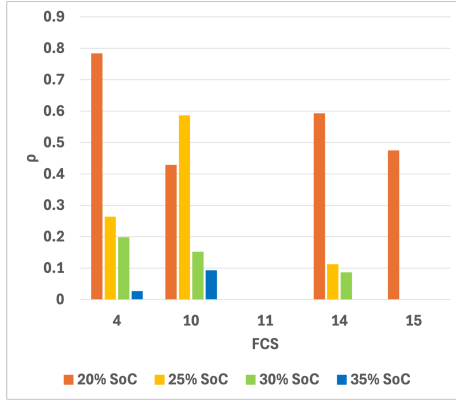
TABLE 7: Charging Station Parameters

Parameter	Value
Number of ports	5
Power	50 kW
Efficiency	100%
Position	0 m to 300 m on lane
Charging Logic	No delay; no in-transit charging

Calibration is conducted by fitting the bounding BPR function to travel-time data obtained from SUMO for these routes under varying evacuation demand levels. For each demand level, the travel time is calculated as the difference between the departure time of the vehicle from the origin node



(a) Box Plot of Evacuation Time vs. Initial SoC



(b) FCS utilization ratio vs. Initial SoC

FIGURE 16: Use Case 4 - Evacuation with different initial SoC.

and the arrival time of the vehicle at the destination node. Figure 23 illustrates the relationship between evacuation flow and travel time. At flow levels below 1,000 vehicles/h, travel time remains relatively stable, fluctuating around a free-flow value of approximately 1.25 times of the free-flow travel time. This indicates that the road capacity is sufficient to handle the demand without significant interaction between vehicles. Beyond this point, the system enters a congested state; the travel time exhibits a sharp upward trend, reflecting the accumulation of delays as the demand approaches and exceeds the link's operational capacity. The red line in this figure shows the BPR bounding function fitted against the normalized travel time data. In the following subsections, we evaluate the proposed evacuation planning approaches in SUMO using the two use cases detailed in Table 8.

E. Use Case 5: Road Traffic Aware Routing in SUMO

In this section, evacuation is assumed to be initiated near Mariposa downtown node 15, affecting nearby nodes 15, 14, 5, 16, 20, and 28, with all evacuees from nodes 15, 14, and 5 routed toward destination node 0 (Westbound),

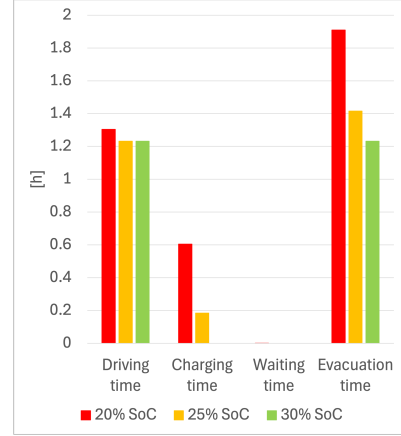


FIGURE 17: Use Case 4 - Evacuation time breakdown for OD pair [20,1] with different initial SoC.

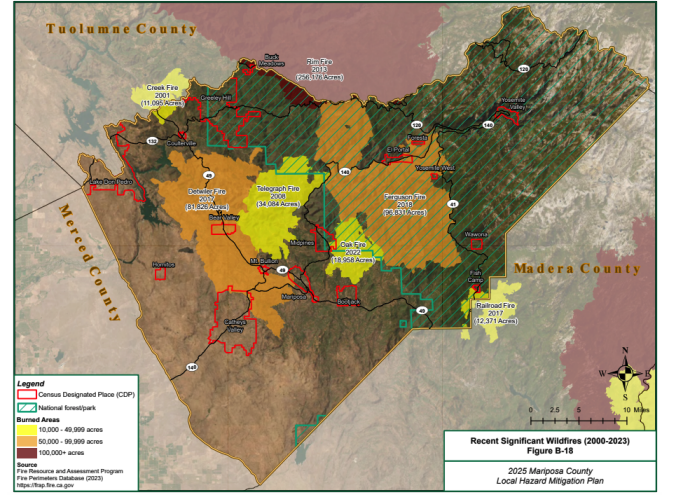


FIGURE 18: Overview of the Mariposa county, California (USA), including main roads and recent wildfires. (Adapted from Mariposa's Local Hazard Mitigation Plan [51]).

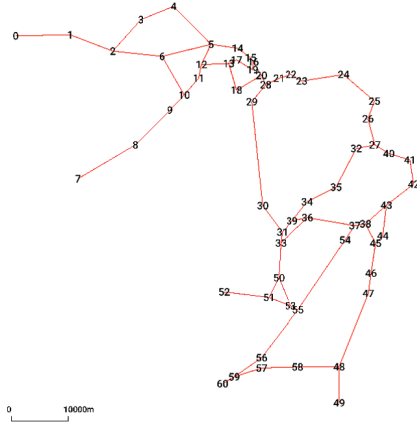
and evacuees from nodes 16, 20, and 28 routed toward destination node 49 (Southbound). Evacuation demand levels of 100 vehicles/hour/OD and 1000 vehicles/hour/OD are evaluated. It is assumed that all vehicles begin the evacuation with a full SoC, requiring no en-route charging to reach their destinations.

1) Performance Comparison Under Varying Evacuation Demand Levels

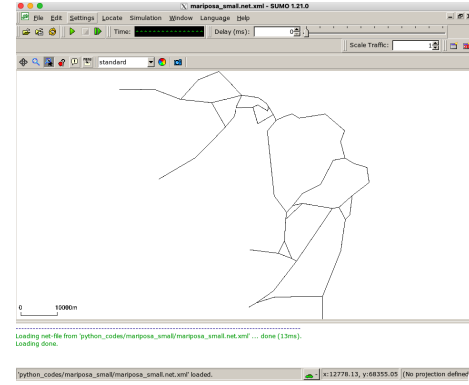
Figure 24 compares the average travel time per vehicle for southbound and westbound evacuations under low and high evacuation demand levels. Under low demand (100 vehicles/hour/OD), all three planning methods converge on identical routing solutions, directing vehicles along the same shortest paths. Consequently, as shown in Figure 24a, these strategies exhibit nearly indistinguishable performance;

TABLE 8: SUMO Simulation Use Cases and Parameters

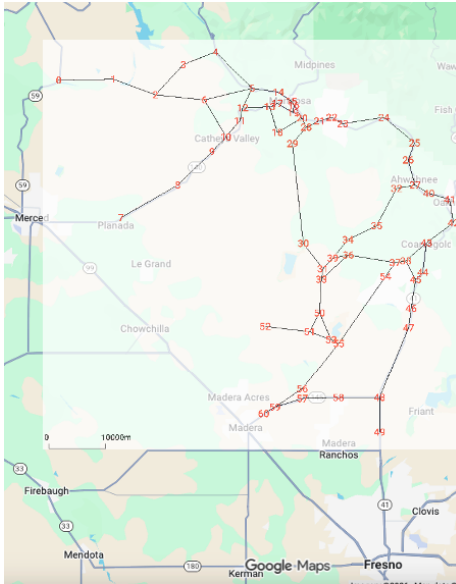
	Use Case	Methods	Num of OD Pairs	Num of ODs needing charge	SoC ₀	Flow per OD	FCS Num	FCS Power	Num of FCS Ports	MCS Num	MCS Power	Num of MCS Ports	Map Scale	β
5a/5b	Road Traffic Aware Routing	UEP/IEP/JEP	6	0	100% (no stop)	100 cars/h; 1000 cars/h	-	-	-	-	-	-	1x	0
6	Charging Traffic Aware Routing with MCS	UEP/IEP/JEP	3	3	20%	50 cars/h	2	50 kW	5	1	50 kW	5	1x	0



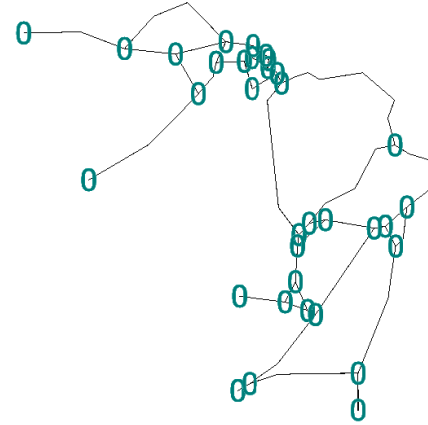
(a) Abstracted network topology



(a) Screenshot of the Mariposa network implemented in SUMO



(b) Network overlaid on geographic map



(b) Nodes with Traffic lights

FIGURE 20: Simplified Mariposa traffic network in SUMO.

FIGURE 19: Simplified Mariposa traffic network used in the evacuation study.

average travel times remain close to free-flow values at approximately 1.9 hours for Southbound and 0.9 hours for Westbound evacuations. This lack of variance indicates that under uncongested conditions, routing coordination offers limited additional benefit.

In contrast, under high demand (1000 vehicles/hour/OD), shown in Figure 24b, clear performance differences emerge. The average travel times increase significantly in both directions. Among the three strategies, the UEP travel time spikes to over 3 hours Southbound and 1.7 hours Westbound. JEP and IEP successfully mitigate these increases. In the Southbound direction, JEP offers a 15.8% benefit by reducing travel time to 2.55 hours, while IEP provides an 11.2% reduction to 2.69 hours. The benefits are even more pronounced for Westbound traffic: compared to UEP, JEP reduces travel time to 1.44 hours (a 16.3% reduction),

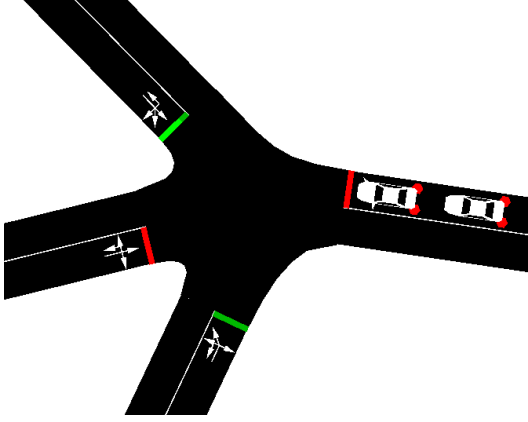


FIGURE 21: Vehicles stopped at node 5 due to the red traffic light.

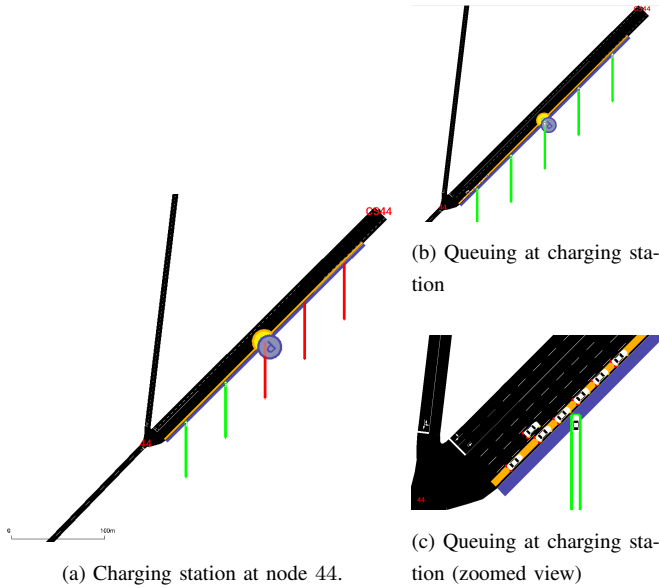


FIGURE 22: Charging station modeling and queuing behavior in SUMO (node 44).

and IEP achieves the highest directional efficiency with a 16.9% reduction to 1.43 hours. Overall, these results indicate that, while all strategies perform similarly under light traffic, the advantages of coordinated evacuation planning become significant as demand increases, leading to meaningful reductions in average evacuation time under high-demand scenarios.

2) Performance Comparison Under High Demand

Figures 25 and 26 summarize the routing behavior and performance of the evacuation strategies under high demand (1000 vehicles/hour/OD).

Figure 25 illustrates the routing assignments for southbound and westbound evacuations. Under high demand, UEP

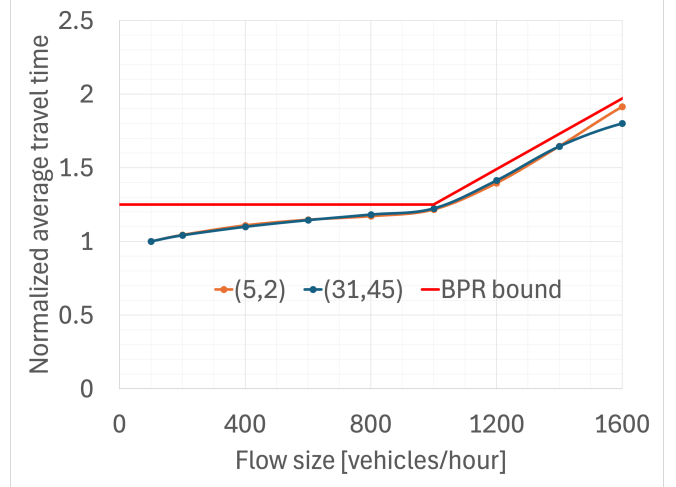
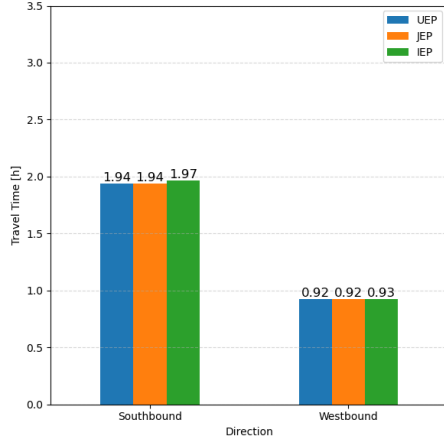


FIGURE 23: BPR bounding function $\hat{t}_{ij}$ (red line) fitted against travel time data from SUMO (blue and orange lines).

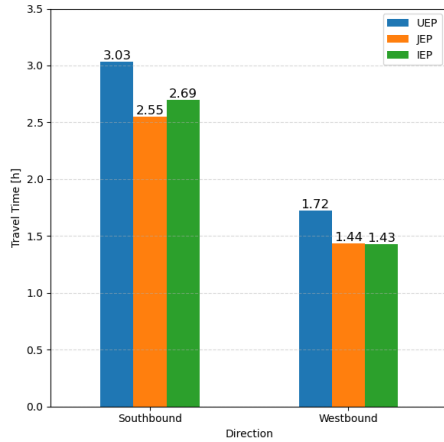
concentrates traffic on a small number of shortest paths, leading to pronounced route overlap and congestion. JEP distributes flow more evenly by diversifying routes. IEP keeps the high-risk OD pairs (16, 49) and (15, 0) on the shortest path, while directing less-at-risk OD pairs to take the detour.

Figure 26 presents the corresponding performance comparison in terms of average travel time and variability for each OD pair. Across all OD pairs, UEP consistently yields the highest average travel times and the greatest variability. This is particularly evident in the southbound high-risk OD pair (16, 49), where the average travel time is 3.38 hours and the maximum time nears 3.9 hours. While UEP struggles with unmanaged congestion, JEP generally improves upon these results by reducing average travel times across all risk levels. However, JEP is notably less effective for high-risk OD pairs when compared to IEP. This disparity arises because IEP specifically prioritizes high-risk OD pairs by routing them along the shortest paths. For instance, the southbound high-risk OD pair (16, 49), shifting from UEP (3.38h) to JEP (2.81h) yields a 16.8% time saving. IEP (2.69h) further increases these savings to 20%. This trend is also reflected in the results for the westbound high-risk OD pair (15, 0), where IEP consistently maintains the lowest average travel time (1.75h), outperforming both UEP (2.10h) and JEP (1.82h).

The improvements offered by IEP in high-risk OD pairs come at the expense of increased evacuation times for certain intermediate pairs, which can be directly evaluated using the Worst-Off-Zone Delay Metric ($\Delta T = T_{IEP} - T_{JEP}$). As shown in Figure 26, for the Westbound evacuation under high demand (1000 vehicles/hour/OD), the medium-risk OD pair (14, 0) is assigned lower priority under IEP to preserve shortest-path capacity for the high-risk pair (15, 0). Consequently, pair (14, 0) experiences the maximum delay



(a) Average Travel Time per Vehicle with 100 vehicles/hour/OD demand size.

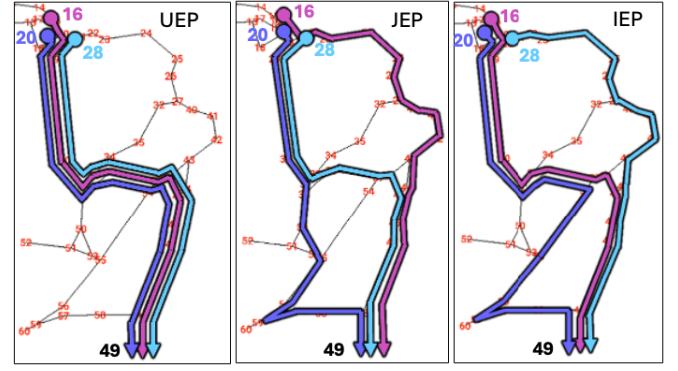


(b) Average Travel Time per Vehicle with 1000 vehicles/hour/OD demand size.

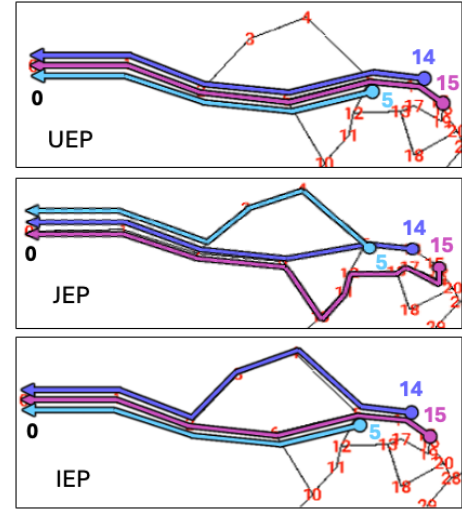
FIGURE 24: Use case 5 - Southbound and Westbound Average Travel Time per Vehicle under different evacuation demand sizes.

penalty in the network, with an average travel time of 1.44 hours under IEP compared to 1.31 hours under JEP ($\Delta T = +0.13$ h = +7.8 min, a 9.9% delay penalty). In contrast, the furthest low-risk pair (5,0) achieves a slight travel-time reduction (1.12 h under IEP vs. 1.18 h under JEP) as upstream bottlenecks clear earlier. While the 9.9% performance penalty is localized to intermediate corridor (14,0), IEP simultaneously achieves a 16.8% to 20.4% reduction in travel time for high-risk Southbound (16,49) and Westbound (15,0) pairs. This concrete metric demonstrates that IEP accepts a ΔT penalty on intermediate routes to maximize life-safety clearance for populations in immediate danger.

Furthermore, the behavioral patterns observed in this high-demand SUMO simulation are consistent with the low-fidelity results presented in Section C. This alignment con-



(a) Southbound evacuation routes.



(b) Westbound evacuation routes.

FIGURE 25: Use case 5 - Routing assignments under high demand (1000 vehicles/hour/OD) for (a) Southbound and (b) Westbound evacuations.

firms that the relative performance and strategic trade-offs of these evacuation models remain robust across varying modeling environments and levels of granularity.

F. Use Case 6: Charging Traffic Aware Routing with MCS in SUMO

In this section, an evacuation scenario similar to the southbound direction in Section 1 is evaluated. Evacuees from nodes 16, 20, and 28 are routed toward destination node 49 with an evacuation demand level of 50 vehicles/hour/OD. All vehicles are assigned a low initial SoC of 20%; consequently, all OD pairs require en-route charging to reach the destination. This case also assumes that charging stations near downtown areas are offline due to the hazard (wildfire); the only available FCSs are located at node 41 (representing a cluster of charging stations in Oakhurst, CA) and node 44 (Chukchansi Crossing Fuel Station and Travel Center).

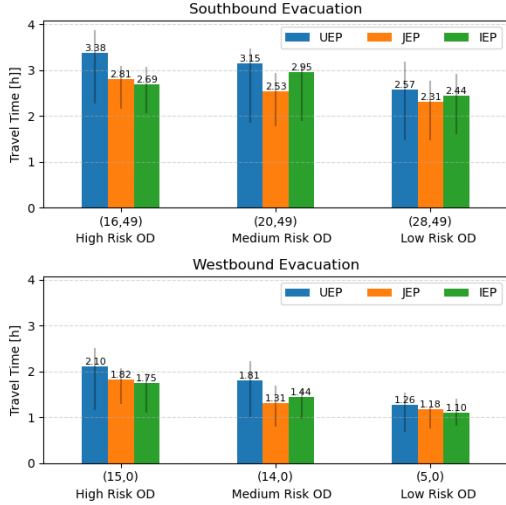


FIGURE 26: Use case 5 - Average evacuation times per OD pair and variability (min/max) under high demand (1000 vehicles/hour/OD).

For the JEP and IEP scenarios, an additional MCS can be deployed to assist with the evacuation process.

Figure 27 illustrates the routes and charging plans generated by the different planning approaches. In the UEP case, all three OD pairs follow the same trajectories as the full-SoC case shown in Figure 25a, with the addition of a charging stop at FCS 44 via link $45 \rightarrow 44$. For JEP and IEP, the routing for all three OD pairs are identical. An additional MCS is deployed at Node 28 or 29 specifically to serve OD pair (20, 49), while the other two OD pairs utilize FCSs located at Nodes 41 and 44.

Figure 28 shows the average and variation of evacuation time for this case using the three planning approaches. The plot reveals that UEP consistently results in the highest mean evacuation times, ranging from 2.35 to 2.54 hours across all risk categories. The high variance and larger error bars observed in the UEP case suggest significant congestion bottlenecks. In contrast, both JEP and IEP substantially improve system efficiency, reducing mean evacuation times by approximately 18% to 26%. The most notable improvement occurs for the Medium Risk OD pair (20, 49), where the evacuation time reduced from 2.39 hours in the UEP case to 1.76 hours in the IEP case. Figure 29 clarifies the drivers behind these performance gains when switching from UEP to JEP or IEP. The actual time spent charging (shown in Figure 29a) remains relatively stable across all scenarios, averaging under 10 minutes. However, Figure 29b shows that the primary bottleneck in the UEP scenario is the extreme congestion at FCS 44, where average waiting times exceed 30 minutes. Simultaneously, FCS 41 remains completely unutilized under UEP, highlighting the inefficiency of uncoordinated routing. JEP and IEP successfully mitigate this by distributing demand across both fixed stations (FCS 41

and 44) and the Mobile Charging Stations (MCS 28 or 29), which confirms that evacuation delays are primarily driven by station queueing rather than charging speeds. By incorporating MCS at nodes 28 or 29, JEP and IEP approaches drastically reduce the total time spent at charging stations. Specifically, JEP and IEP achieve an approximate 79% to 88% reduction in time spent at charging stations compared to the UEP bottleneck. This load-balancing effect effectively stabilizes the network, resulting in significantly lower total evacuation times observed in Figure 28. The observed results are consistent with the results presented in Section D.

G. Computational Scalability and Performance Comparison

To evaluate the computational scalability of the proposed planning methods, Figure 30 and Table 9 illustrates the solver performance for the UEP, IEP, and JEP strategies across three distinct evaluation scenarios. All optimization problems were implemented in Python and solved using the Gurobi Optimizer (version 11.0.1) [42]. Computations were executed on an Apple M2 processor with 8 physical cores, 8 logical processors, and 32 GB RAM operating under macOS (Darwin 23.3.0). The solver was configured to utilize up to 8 parallel threads with default presolve and branch-and-bound settings. To balance solution quality and computational performance on complex instances, a relative mixed integer programming (MIP) gap tolerance of 10% was specified. In the less computationally demanding scenarios without charging decisions (Use Cases 5a and 5b), both methods show comparable optimality gaps. JEP achieves a faster overall solve time (0.07 seconds), while IEP (0.17 and 0.18 seconds, respectively) offers a computational advantage when planning for high-risk groups in the first two iterations (0.01 to 0.03 seconds per iteration). However, this dynamic shifts significantly as problem complexity and network constraints increase. In Use Case 6, the monolithic formulation of JEP causes a combinatorial explosion, resulting in a severe solve-time spike to 185.88 seconds alongside a substantially higher optimality gap (9.98%), whereas IEP successfully mitigates this computational bottleneck by calculating the complete multi-group evacuation plan in just 0.37 seconds distributed across three iterations with lower, well-controlled gaps.

VI. Conclusion

This paper presents a comprehensive optimization framework for ZEV evacuation that integrates routing, charging, traffic congestion, and MCS deployment. Formulated as a MILP, the model captures real-world constraints to support resilient evacuation strategies. We also proposed an iterative evacuation planning approach (IEP) that effectively prioritizes high-risk zones based on hazard proximity through a recursive optimization process. The IEP provides emergency planners with a flexible tool to build the evacuation process

TABLE 9: Optimality Gap (%) Comparison

Method	Iteration	Use Case 5a	Use Case 5b	Use Case 6
IEP	iter1	0.00%	0.00%	0.00%
	iter2	8.14%	8.14%	0.00%
	iter3	0.00%	0.00%	0.00%
	iter4	9.73%	8.93%	—
	iter5	5.71%	6.76%	—
	iter6	0.00%	0.00%	—
JEP	Final	0.00%	7.35%	9.98%

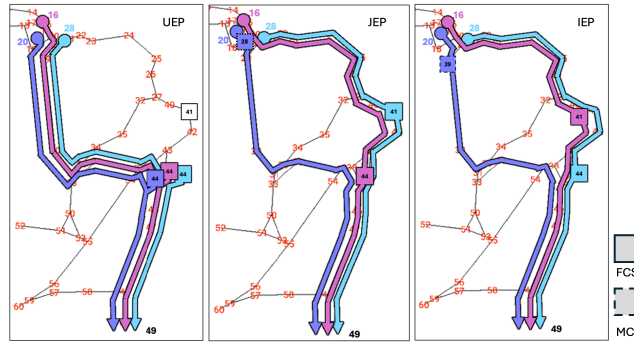


FIGURE 27: Use Case 6 - Routing and charging plan with different planning approaches.

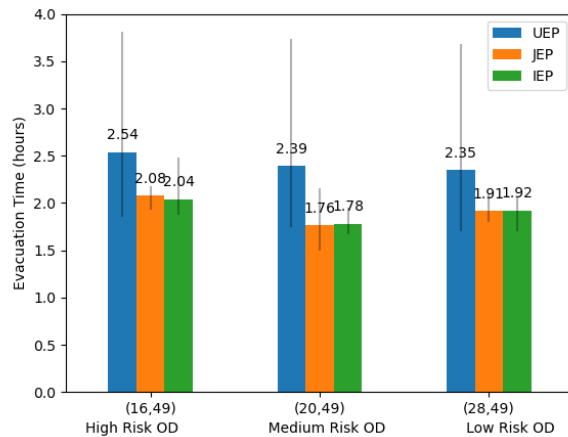
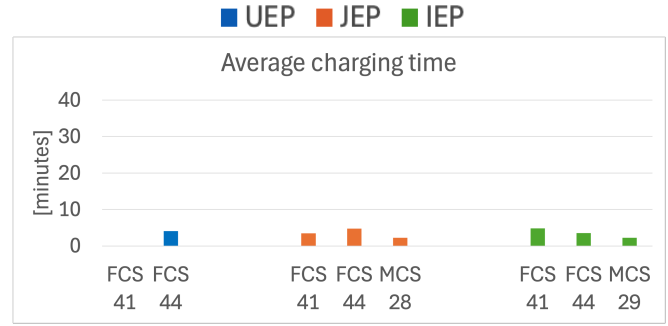
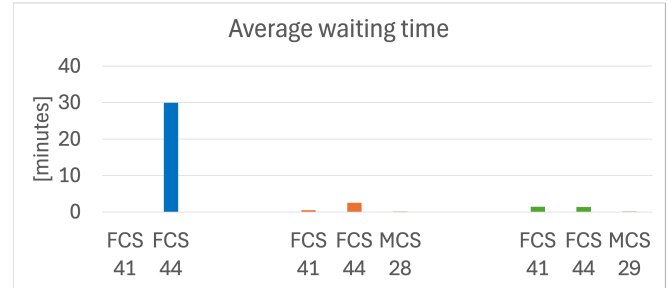


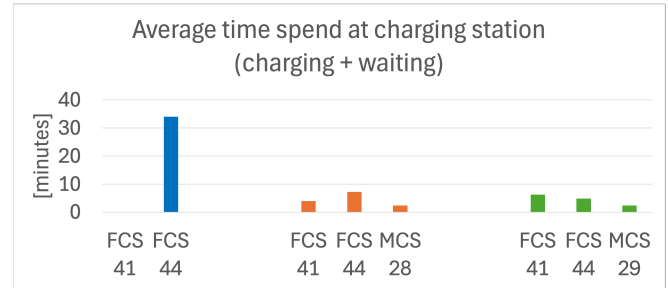
FIGURE 28: Use Case 6 - Evacuation time with different planning approaches.



(a) Average charging time.



(b) Average waiting time.



(c) Average time spent at charging station.

FIGURE 29: Use case 6 - Average time spent at charging stations under different planning methods.

step-by-step, starting with the neighbors closer to danger, and then progressing toward lower-risk regions. We observed that the IEP approach may reduce computational time of the optimization process, especially for larger and more complex evacuation problems.

We evaluated the proposed optimization methods using both a low-fidelity traffic model on the Sioux Falls network and the high-fidelity SUMO simulator on a realistic Mariposa network. Compared to uncoordinated evacuation (UEP), the proposed optimization approaches (IEP, JEP) significantly reduces evacuation times, improves charger utilization, and yields more balanced traffic and charging loads across the networks. The results also underscore the critical role of MCS when fixed infrastructure becomes unavailable due to hazard-induced power outages. Additionally, sensitivity analyses show that access to charging infrastructure becomes

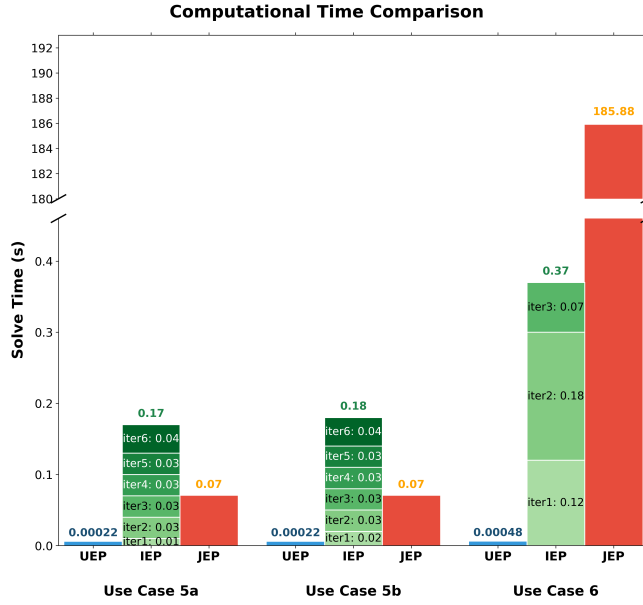


FIGURE 30: Computational Time Comparison

be re-solved periodically in a rolling-horizon fashion to re-plan evacuation routes and re-deploy MCSs as dynamic traffic demand, hazard boundaries, and available MCS resources evolve. This is another research avenue we will explore in future works.

increasingly important as vehicles' initial SoC decreases and evacuation distances increase. These findings offer valuable insights for transportation planners and emergency response agencies developing ZEV-focused evacuation strategies in hazard-prone regions.

While the proposed framework provides valuable insights for ZEV evacuation planning, it has several limitations. First, the model is based on a steady-state traffic flow assumption, which does not capture the dynamic and time-varying nature of real-world traffic during emergency evacuations. Specifically, this equilibrium approach likely underestimates transient congestion spikes at sudden bottlenecks and may overestimate total system clearance times due to the omission of spontaneous, dynamic rerouting by drivers.

Future work will also explore i) introducing fractional decision variables to model fleets subdivided into smaller units following different routes, and ii) relaxing the loop-free routing constraint to accommodate dynamic detours and alternative routing behaviors around sudden hazards. Such relaxations could provide additional insights while leveraging the computational efficiency of continuous optimization to further improve the practical applicability and scalability of the framework. Additionally, the current model assumes uniform MCS charging power and constant capacity. In practice, dynamic operational factors, such as generator load variations, fuel logistics, and local grid constraints can cause real-time power availability to fluctuate during a disaster. Therefore, incorporating dynamic power availability in our optimization framework represents a key avenue for future research.

Finally, while the optimization formulation presented in our manuscript provides a single-stage deployment plan over a target horizon, the model is fundamentally modular and can

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Appendix

A. Proof of Proposition 1

Proof:

Let us divide $\mathcal{D}_f$ into two sub-domain: i) $\mathcal{D}_f^A = [0, f_{ij}^*]$ and ii) $\mathcal{D}_f^B = (f_{ij}^*, f_{ij}^* + \Delta f_{ij}^{max}]$. In the first sub-domain ($\mathcal{D}_f^A$) we obtain $t_{ij}(f_{ij}) \leq \hat{t}_{ij}^{ff} = t_{ij}(f_{ij}^*)$ for all $f_{ij} \in \mathcal{D}_f^A$, where the inequality stems from the fact that $t_{ij}(f_{ij})$ is an increasing function. This demonstrates that $t_{ij}(f_{ij}) \leq \hat{t}_{ij}(\Delta f_{ij})$ in the first sub-domain. In the second sub-domain, we introduce the linear function $g_{ij}(f_{ij}) = a_{ij} + b_{ij}f_{ij}$ and compute a_{ij}, b_{ij} such that $g_{ij}(f_{ij}^*) = t_{ij}(f_{ij}^*)$ and $g_{ij}(f_{ij}^* + \Delta f_{ij}^{max}) = t_{ij}(f_{ij}^* + \Delta f_{ij}^{max})$. This yields

$$\begin{aligned} \Delta f_{ij}^{max} &= t_{ij}(f_{ij}^* + \Delta f_{ij}^{max}). \text{ This yields} \\ a_{ij} &= t_{ij}(f_{ij}^*) - k_{ij}f_{ij}^*, \quad b_{ij} = k_{ij} \end{aligned} \quad (83)$$

Note that $b_{ij} > 0$ since t_{ij} is a monotonically increasing function. Since $t_{ij}(f_{ij})$ is a convex function (for $b_2 \geq 1$), the chord of the function t_{ij} (which is given by g_{ij}) is an "upper bound" for the graph of t_{ij} along a line segment connecting two arbitrary points in the domain [53]. We set f_{ij}^* and $f_{ij}^* + \Delta f_{ij}^{max}$ as the end-points of the line segment, yielding:

$$t_{ij}(f_{ij}) \leq g_{ij}(f_{ij}), \quad \forall f_{ij} \in \mathcal{D}_f^B \quad (84)$$

Additionally, the vehicle flow is bounded by (83). Thus:

$$g_{ij}(f_{ij}) \leq g_{ij}(f_{ij}^* + \Delta f_{ij}), \quad \forall f_{ij} \in \mathcal{D}_f^B \quad (85)$$

$$= a_{ij} + b_{ij}f_{ij}^* + b_{ij}\Delta f_{ij} \quad (86)$$

$$= t_{ij}(f_{ij}^*) + k_{ij}\Delta f_{ij} \quad (87)$$

$$= \hat{t}_{ij}^{ff}(\Delta f_{ij}) \quad (88)$$

for all $f_{ij} \in \mathcal{D}_f^B$, where the first inequality stems from the fact that g_{ij} is an increasing function. ■

B. Proof of Proposition 2

Proof:

(Optimality) Let us consider the cost function of $\mathbb{P}^{JEP}$, which has five terms. The minimum value of the last three is zero, which is obtained by fixing $y_j^m = 0$ (no stop for charging), $\Delta e_j^m = 0$ (no re-charge energy), and $z_i = 0$ (no mobile charging deployment). The CACF solution $\tilde{x}_{ij}^m, \Delta \tilde{f}_{ij}^m$ minimizes the first two terms due to the formulation of $\tilde{\mathbb{P}}^{JEP}$, while (76) minimizes the last three.

(Feasibility) We need to check that i) $\hat{\chi}$ exists and that ii) it fulfills the constraints of $\mathbb{P}^{JEP}$. The existence of $\hat{\chi}$ can be established due to Assumption 1 (which ensure that there is always a path connecting o_m and d_m). Regarding the constraints of $\mathbb{P}^{JEP}$. The transportation constraints (53) to (55) are automatically satisfied by $\tilde{x}_{ij}^m, \Delta \tilde{f}_{ij}^m$ (since they are part of $\mathbb{P}^{JEP}$). The mobile charging constraints are also fulfilled (as long as $n_{MCS} \geq 0$). The critical constraint is $e_j^m \geq 0$. To check this, it is convenient to re-write (58) and (59) as

$$e_j^m = e_i^m - \eta_{ij}^m \tilde{x}_{ij}^m \geq 0, \quad (i, j) \in \mathcal{L}_T \quad (89)$$

which makes use of the fact that $\Delta \tilde{e}_j^m = 0, j \in \mathcal{V}_T$. Let us consider the evacuation path induced by $\tilde{x}_{ij}^m$, which is denoted as $(l_{m,1}, l_{m,2}, \dots, l_{m,W_m})$, where W_m is the number of road segments used by the vehicle fleet m during the evacuation². The available energy after visiting the node $l_{m,k+1}$ can be written as:

$$e_{l_{m,k+1}}^m = e_{l_{m,k}}^m - \eta_{l_{m,k}, l_{m,k+1}}^m \tilde{x}_{l_{m,k}, l_{m,k+1}}^m \quad (90)$$

$$= e_{start}^m - \sum_{h=1}^{k+1} \eta_{l_{m,h}, l_{m,h+1}}^m \tilde{x}_{l_{m,h}, l_{m,h+1}}^m \geq 0 \quad (91)$$

²Note that $l_{m,1} = o_m$ and $l_{m,W_m} = d_m$.

for $k = 1, 2, \dots, W_m - 1$. This energy is non-negative if the following inequality holds:

$$e_{start}^m \geq \sum_{h=1}^{k+1} \eta_{l_m, h, l_m, h+1}^m \tilde{x}_{l_m, h, l_m, h+1}^m \quad (92)$$

for $k = 1, 2, \dots, W_m - 1$. It can be shown that:

$$e_{start}^m \geq \sum_{h=1}^{W_m} \eta_{l_m, h, l_m, h+1}^m \tilde{x}_{l_m, h, l_m, h+1}^m \quad (93)$$

$$\geq \sum_{h=1}^{k+1} \eta_{l_m, h, l_m, h+1}^m \tilde{x}_{l_m, h, l_m, h+1}^m \quad (94)$$

for $k = 1, 2, \dots, W_m - 1$. The first inequality is valid due to assumption (77); the second inequality stems from property of finite series ($\sum_{h=1}^{W_m} |q_h| \geq \sum_{h=1}^{k+1} |q_h|$ for any $1 \leq k \leq W_m - 1$, where $|q_h| = \eta_{l_m, h, l_m, h+1}^m \tilde{x}_{l_m, h, l_m, h+1}^m$). As a result (92) is fulfilled and all e_j^m are non-negative, demonstrating feasibility of $\hat{\chi}$. ■

C. Proof of Proposition 3

Proof:

We need to show feasibility of the candidate solution. $\tilde{x}_{ij}^m, \Delta f_{ij}$ fulfills the transportation constraints since they were incorporated into $\hat{\mathbb{P}}^{JEP}$. Since we assumed that the origins have chargers (i), we can inject enough energy (80) in the vehicles to reach the destination: for all $m \in \mathcal{M}$. This energy can be stored in the vehicle due to assumption ii). As a result, we do not need to stop in charger along the evacuation route ($y_j^m = 0, \Delta e_j^m = 0, j \in \mathcal{V}_{CS} \setminus \{o_m\}$) neither deploy mobile chargers ($z_i = 0, i \in \mathcal{V}_{MCS}$). ■

D. Auxiliary optimization Problem

$$\hat{\mathbb{P}}^{JEP} : \min \frac{1}{M} \sum_{m \in \mathcal{M}} \left(\sum_{(i,j) \in \mathcal{L}_T} \hat{t}_{ij}^{ff} x_{ij}^m + k_{ij} \Delta f_{ij} \right) \quad (95)$$

s.t. $m \in \mathcal{M}$

$$\sum_{i \in \mathcal{V}^+(k)} x_{ik}^m = \sum_{j \in \mathcal{V}^-(k)} x_{kj}^m + \mathbf{1}_{k=d_m} - \mathbf{1}_{k=o_m}, \quad k \in \mathcal{V}_T \quad (96)$$

$$x_{ij}^m \in \{0, 1\}, \quad (i, j) \in \mathcal{L}_T \quad (97)$$

$$\sum_{i \in \mathcal{V}^+(k)} x_{ik}^m \leq 1, \quad k \in \mathcal{V}_T \quad (98)$$

$$f_{ij} = \beta_{ij} + \sum_{m \in \mathcal{M}} \alpha_m x_{ij}^m \leq f_{ij}^* + \Delta f_{ij}, \quad \Delta f_{ij} \geq 0 \quad (99)$$

E. Quantification of Energy Model Conservatism (Assumption 4)

To evaluate the conservatism of Assumption 4, we analyze empirical vehicle efficiency distributions from the EV Database [54], [55], as illustrated in Figure 31. As shown in Figure 31, setting a uniform rate of $e_{worst} = 0.25$ kWh/km

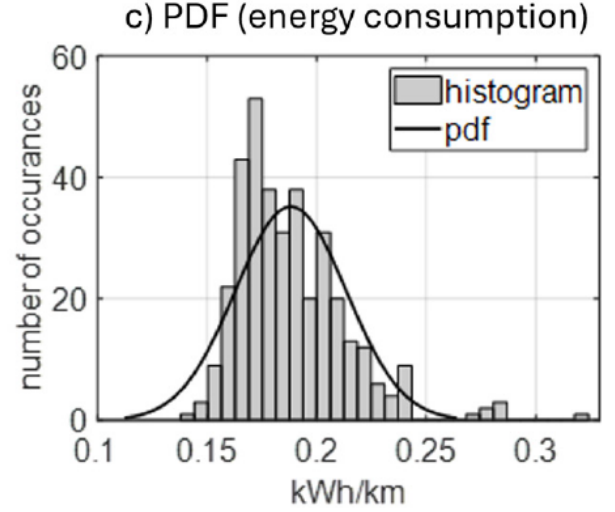


FIGURE 31: EV Energy consumption distribution [54], [55]

bounds approximately 98% of commercially available EV models under standard operating conditions. Comparing this upper bound to the empirical fleet mean ($\bar{e} \approx 0.19$ kWh/km) yields:

$$\begin{aligned} \text{Conservatism Margin} &= 1 - \frac{\bar{e}}{e_{worst}} \\ &= 1 - \frac{0.19 \text{ kWh/km}}{0.25 \text{ kWh/km}} \\ &= 24\% \end{aligned} \quad (100)$$

This demonstrates that Assumption 4 incorporates an approximate 24% operational energy safety buffer to guarantee zero stranded vehicles during evacuations. Crucially, while overall energy demand drops by 24% under the subgroup split, Assumption 4 provides a safe operational upper bound that ensures all vehicles complete their routes without altering the underlying spatial network constraints.